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# Experimental and kinetic modeling study of $\alpha$-methyl-naphthalene pyrolysis: Part II. PAH formation

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ABSTRACT

$\alpha$-Methyl-naphthalene plays an important role as a functional material in petrochemical industries and as a precursor of soot particles. The formation chemistry of polycyclic aromatic hydrocarbons (PAHs) from $\alpha$-methyl-naphthalene, therefore, warrants detailed investigations. In this work, we studied PAH formation from its pyrolysis using experiments and kinetic models. Flow reactor pyrolytic experiments at low and atmospheric pressures (30 and 760 Torr) were performed using synchrotron vacuum ultraviolet photoionization molecular beam mass spectrometry (SVUV-PI-MBMS). A kinetic model was then developed to predict PAH formation from $\alpha$-methyl-naphthalene. According to the kinetic analysis of the proposed model, naphth-1-yl-methyl, benzo-fulvenallene, and benzo-fulvenallenyl are three critical intermediates in the formation of large PAHs. Other than the traditional H-abstraction acetylene-/vinylacetylene-addition mechanisms, three prototypical PAH formation pathways are identified in $\alpha$-methyl-naphthalene pyrolysis: 1) addition and cyclization reactions of naphth-1-yl-methyl and naphth-1-yl radicals; 2) recombination of resonance stabilized radicals (indenyl, benzo-fulvenallenyl, phenalenyl, etc.) and the subsequent ring expansion reactions; 3) sequential propargyl addition reactions.



## 1. Introduction

$\alpha$-Methyl-naphthalene ($\alpha A_2CH_3$) is a critical intermediate in coal coking process [1,2], coal-derived liquids [3], industrial synthesis of light aromatic hydrocarbons [4] and highly graphitizable pitches [5], and the formation of polycyclic aromatic hydrocarbons (PAHs) and particular matter in combustion [6]. Thermal chemistry of $\alpha$-methyl-naphthalene is thus of high relevance in understanding and predicting the formation of PAHs. Previous experimental investigations on $\alpha$-methyl-naphthalene reported speciation information under pyrolysis and oxidation conditions at various pressures and temperatures [7-11]. Two simple PAH models were developed to unveil the aromatic growth processes based on $\alpha$-methyl-naphthalene [7,10]. Among all previous studies, only Mati et al. [7] reported the chemistry of $\alpha$-methyl-naphthalene at combustion-relevant conditions. They measured small species in $\alpha$-methyl-naphthalene oxidation in a jet-stirred reactor at 800 – 1421 K and 1 – 13 atm by gas chromatography-mass spectrometry (GC–MS), and developed a kinetic model by correlating the chemistry of $\alpha$-methyl-naphthalene to that of naphthalene. Since their model was primarily targeted at the oxidation of $\alpha$-methyl-naphthalene, it did not include much information on PAH formation chemistry. Other literature studies on $\alpha$-methyl-naphthalene pyrolysis have been performed at supercritical conditions (critical temperature: 499 °C, critical pressure: 36 atm) [8-11]. These works were carried out at similar conditions (380 – 585 °C and 40 – 110 atm) and measured the products by GC–MS, high-pressure liquid chromatography (HPLC), MS, and UV absorbance spectroscopy [8,9,11]. Bounaceur et al. [10] built up a pyrolytic model for $\alpha$-methyl-naphthalene to describe its chemistry at the supercritical state. Their model focused only on the radical combination reactions heading for molecular weight growth at supercritical conditions, and neglected the decomposition of naphth-1-yl-methyl and methyl-naphthyl radicals as well as the contribution of secondary fuel-derived radicals to PAH formation.

Previous PAH models [12-18] did not include aromatic growth pathways from $\alpha$-methyl-naphthalene, and primarily converted it to naphth-1-yl radical for further reactions. The reactions of

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naphth-1-yl-methyl radical were generally analogized to those of benzyl without considering the particular molecular structure of naphth-1-yl-methyl radical. Naphth-2-yl-methyl radical is quite similar to benzyl, but naphth-1-yl-methyl is not due to the *zigzag* periphery next to its radical site. Specific PAH growth routes from naphth-1-yl-methyl, therefore, should be considered in PAH kinetic studies. Methyl-naphthyl, another primary radical of $\alpha$-methyl-naphthalene, may have similar PAH formation reactions as naphthyl. Here, sub-mechanism of naphthalene may be a good reference for the PAH chemistry of methyl-naphthyl. Phenyl addition-cyclization reaction for PAH growth [19] was not widely included in the previous PAH model [17], which may be analogized to the reactions of naphthyl and methyl-naphthyl. Our previous study on $\alpha$-methyl-naphthalene decomposition showed that benzo-fulvenallene ($C_9H_6CCH_2$) and benzo-fulvenallenyl ($C_9H_6C_2H$) are two critical pyrolytic intermediates (SMM8). Considering the contribution of fulvenallenyl ($C_7H_5$) radical in PAH formation [20,21], benzo-fulvenallenyl may yield large PAHs via similar chain reactions of resonance stabilized radicals.

In accordance with the current knowledge of the PAH chemistry of $\alpha$-methyl-naphthalene, a comprehensive experimental and modeling investigation is required to identify the specific PAHs and their relative precursors. In this study, detailed speciation data of $\alpha$-methyl-naphthalene pyrolysis are measured in a flow reactor using synchrotron vacuum ultraviolet photoionization molecular beam mass spectrometry (SVUV-PI-MBMS) [22], with particular attention to PAHs with more than four aromatic rings. The obtained speciation data are used to develop a new kinetic model for $\alpha$-methyl-naphthalene with updated PAH sub-mechanism based on our previous indene model [17]. A major improvement in the present model is the inclusion of PAH pathways for $C_{11}$ species, such as naphth-1-yl-methyl, methyl-naphthyl, and benzo-fulvenallenyl radicals, which directly correlate the monocyclic and bicyclic aromatic species with PAHs larger than tetracyclic species. Important prototypical reaction schemes are developed to improve our general understanding of the chemistry of PAH and soot.

## 2. Experimental details

$\alpha$-Methyl-naphthalene pyrolysis was studied by synchrotron vacuum ultra-violet photoionization molecular beam mass spectrometry (SVUV-PI-MBMS). The experiment was performed in a flow reactor at an undulator-based VUV beamline (BL03U) of Hefei Light Source (HLS II), China. Details of the experiments were introduced in the Part I of this study, including the flow reactor facility, experimental conditions, data acquisition and evaluation. The measured reactor temperature profiles and species mole fractions are given in SMM1. The uncertainty of SVUV-PI-MBMS measurements comes primarily from the sampling process and the photoionization cross-sections (PICSs) of the measured species [17,23]. Generally, the measurement uncertainty is $\pm$10% for species with direct PICS measurement, $\pm$25% for stable species with PICS available in the literature [24,25] or database [26], and a factor of 2 for free radicals and species with estimated PICS [17]. The mole fractions of active free radicals are usually under-evaluated due to their annihilation in the sampling process. PICSs of all species adopted during the data evaluation process are given in SMM2.

## 3. Model construction

The kinetic model of $\alpha$-methyl-naphthalene can be roughly divided in two parts, decomposition scheme and aromatic growth scheme. The first part was introduced in Part I of this study, and the second part is the main topic of this paper. We aim to provide a detailed picture of PAH pathways in the present model, particularly from bicyclic aromatic species to large molecules with multiple aromatic rings. A number of important reactions, related to naphth-1-yl-methyl and other odd C-number PAH species, have been included in the present model. Rate constants of these reactions are either taken from values in the literature [7,10] or analogized to similar reactions of benzyl [12,13] and fulvenallenyl [17,20]. Detailed kinetic model files are provided in SMM3 – 6, as well as nomenclatures in SMM7.

### 3.1. PAH formation via the recombination of $C_{11}H_9$ radicals

The decomposition of methyl-naphthalene ($\alpha A_2CH_3$) mainly produces two kinds of $C_{11}H_9$ radicals, methyl-naphthyl ($C_{10}H_6CH_3$) and naphth-1-yl-methyl ($\alpha A_2CH_2$). Methyl-naphthyl is not a radical with a specific molecular structure but instead is a lumped representation of all methyl-naphthyl radicals with their radical sites at different C-atoms of benzene rings. Self-combination of naphth-1-yl-methyl (R1) and its combination with methyl-naphthyl (R2) have been proposed in the present model according to previous studies [8,10], yielding dinaphthyl-ethane ($A_2CH_2CH_2A_2$) and methyl-(naphth-1-yl-methyl)-naphthalene ($A_2CH_2A_2CH_3$), respectively. The rate constants of R1 and R2 refer to benzyl recombination [12] and phenyl recombination [27] reactions, respectively.

The C−C bond dissociation of dinaphthyl-ethane (R3) produces naphthyl-ethyl ($A_2CH_2CH_2$) and naphth-1-yl ($A_2$-) radicals, while C−H bond dissociation of $A_2CH_2CH_2A_2$ (R4) produces dinaphthyl-ethyl radical ($A_2CHCH_2A_2$). The rate constants of R3 and R4 are analogized to those of ethyl fission of ethyl-benzene [28] and C−H bond fission of ethane [29], respectively. H-abstraction reactions of dinaphthyl-ethane also produce dinaphthyl-ethyl radical. As shown in Scheme 1, further reactions of dinaphthyl-ethyl radical are similar to those of ethyl radical, yielding dinaphthyl-ethylene and dinaphthyl-acetylene *via* stepwise H-elimination. This part of the present model is taken from the sub-mechanism of ethylene. Meanwhile, $\beta$-C-C bond fission of dinaphthyl-ethyl radical yields vinyl-naphthalene and naphth-1-yl (R5), with its rate coefficient referred to the reaction of dinaphthyl-ethyl radical [28]. H-addition followed by C−C bond fission reactions are considered for dinaphthyl-ethane (R6) and dinaphthyl-ethyl (R7), which produce naphthyl-ethyl, naphthyl, and naphthalene.

$C_{22}H_{15}$ radical derived from dinaphthyl-ethylene has other isomers besides $A_2CCHA_2$. As shown in Scheme 1, Two more $C_{22}H_{15}$ radicals, $A_2CHCHA_2$−2 and $A_2CHCHA_2$−8, are considered in the present model, which are formed *via* H-elimination from the benzene ring. Penta-ring cyclization of $A_2CHCHA_2$−8 may form $A_2R_5HA_2$ radical, which in turn yields naphthyl-acenaphthylene ($A_2R_5A_2$) by releasing an H-atom. The addition of naphth-1-yl to acenaphthylene also makes $A_2R_5HA_2$ radical. Further dehydrogenation of $A_2R_5A_2$ produces benzo-cyclopentatetraphene ($BR_5TEPH$), a large PAH with six aromatic rings. $A_2CHCHA_2$−2 radical, on the other hand, tends toward a six-member-ring cyclization to form picene, which is also a large PAH but with five benzene rings. Similar to the reaction of phenyl and phenylacetylene to form phenanthrene [30,31], the reaction of naphth-1-yl and ethynyl-naphthalene may also contribute to the formation of picene *via* the isomerization reaction between $A_2CCHA_2$ and $A_2CHCHA_2$−2.

$$A_2CH_2CH_2A_2 = \alpha A_2CH_2 + \alpha A_2CH_2 \tag{R1}$$

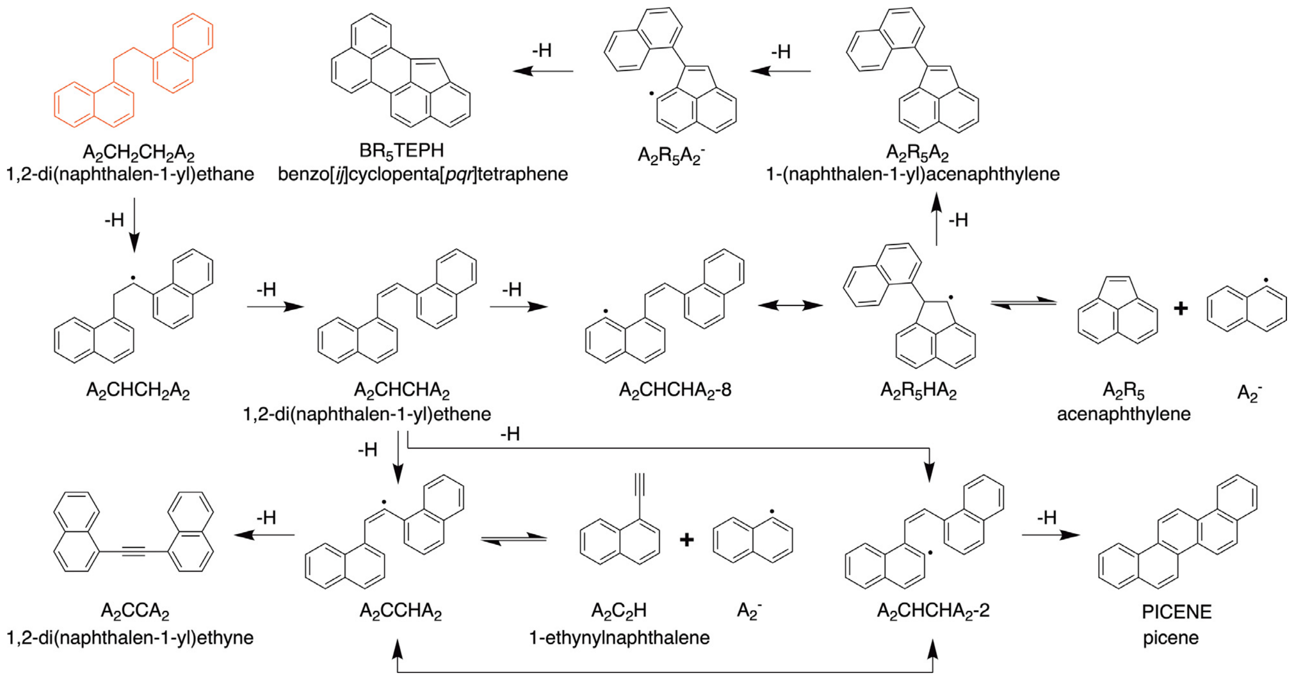
$$C_{10}H_6CH_3 + \alpha A_2CH_2 = A_2CH_2A_2CH_3 \tag{R2}$$

$$A_2CH_2CH_2A_2\ (+M) = A_2CH_2CH_2 + A_2\text{-}\ (+M) \tag{R3}$$

$$A_2CHCH_2A_2 + H\ (+M) = A_2CH_2CH_2A_2\ (+M) \tag{R4}$$

$$A_2CH_2CH_2A_2\ (+M) = A_2CH_2CH_2 + A_2\text{-}\ (+M) \tag{R5}$$

$$A_2CH_2CH_2A_2 + H = A_2CH_2CH_2 + A_2 \tag{R6}$$




**Scheme 1.** Reaction schemes of $\alpha A_2CH_2 + \alpha A_2CH_2$, $A_2$- + $A_2R_5$ and $A_2$- + $A_2C_2H$, as well as the subsequent reaction steps for the formation of picene, benzo-cyclopenta-tetraphene and dinaphthyl-acetylene. One- and two-way arrows indicate dehydrogenation and isomerization steps, respectively; double arrows indicate entrance channels.

$$A_2CHCH_2A_2 + H = A_2CH_2CH_2 + A_2\text{-} \quad (R7)$$

For the further reaction of $A_2CH_2A_2CH_3$, we considered C–H bond fission (R8) and H-abstraction to form $A_2CHA_2CH_3$, C–C bond fission to form (methyl-naphthyl)-methyl ($CH_3A_2CH_2$) and $A_2$- radicals (R9), and H-addition followed by C–C bond fission reactions (R10 – R12). The rate constants of R8 and R10 – R12 are referred to the similar reactions of toluene [32,33]. Reaction R9 is analogized to the reaction of benzyl and phenyl [12]. Dinaphthyl-methane ($A_2CH_2A_2$) is one of the major products from $A_2CH_2A_2CH_3$ decomposition (R12). Another $A_2CH_2A_2$ formation reaction is the combination of naphth-1-yl and naphth-1-yl-methyl radicals (R13). The rate constants of R9 are adopted for this reaction. $A_2CHA_2$, a radical of $A_2CH_2A_2$, forms 7*H*-benzo[*ij*]tetraphene (BiTEPH) *via* H-shift isomerization and cyclization.

$$A_2CH_2A_2CH_3\ (+M) = A_2CHA_2CH_3 + H\ (+M) \quad (R8)$$

$$CH_3A_2CH_2 + A_2\text{-} = A_2CH_2A_2CH_3 \quad (R9)$$

$$A_2CH_2A_2CH_3 + H = \alpha A_2CH_3 + \alpha A_2CH_2 \quad (R10)$$

$$A_2CH_2A_2CH_3 + H = CH_3A_2CH_2 + A_2 \quad (R11)$$

$$A_2CH_2A_2CH_3 + H = A_2CH_2A_2 + CH_3 \quad (R12)$$

$$A_2\text{-} + \alpha A_2CH_2 = A_2CH_2A_2 \quad (R13)$$

### 3.2. PAH pathways from naphth-1-yl-methyl radical

#### 3.2.1. Methyl addition to naphth-1-yl-methyl radical

Methyl addition to naphth-1-yl-methyl forms ethyl-naphthalene which has a similar structure as ethyl-benzene, therefore, a similar reaction scheme to that of ethyl-benzene is adopted in the present model. The step-wise dehydrogenation of ethyl-naphthalene sequentially produces vinyl-naphthalene and ethynyl-naphthalene. The rate constants of these reactions are referred to the previous ethyl-benzene model [28]. Different from phenyl-vinyl radical, naphthyl-vinyl is able to cyclize to acenaphthylene *via* H-elimination. Detailed chemistry of these $C_{12}$ PAHs has been introduced in previous study [34].

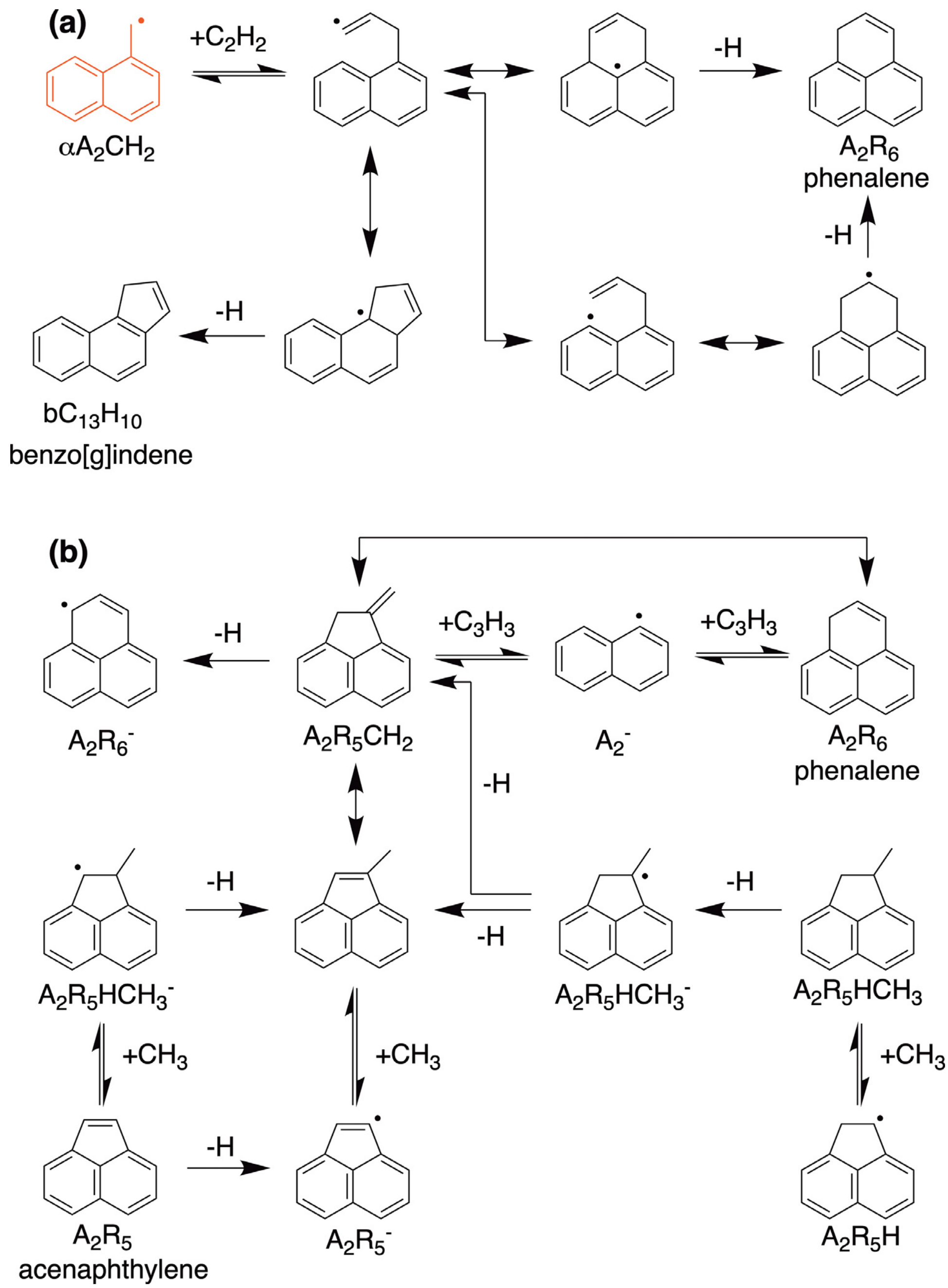
#### 3.2.2. Addition of $C_2$ species to naphth-1-yl-methyl radical

Acetylene addition to benzyl radical has been identified as an important indene formation pathway in previous studies [35]. Similar reactions are also available for naphth-1-yl-methyl (R14), as shown in Scheme 2a. Since the methyl group of naphth-1-yl-methyl is at the $\alpha$-site, the product of this reaction is assumed to be benzo[*g*]indene. However, another pathway is possible for such a cyclization after acetylene addition to naphth-1-yl-methyl (R15), which crosses over the *zigzag* periphery of naphthalene to fuse phenalene ($A_2R_6$). Phenalene may be more favorable than benzo[*g*]indene because the six-member-ring in phenalene has low ring strain energy than the five-member-ring in benzo[*g*]indene. As shown in Scheme 2a, R15 has a similar reaction sequence as benzene formation *via* $C_4H_5 + C_2H_2$. Theoretical calculation results of Miller and Klippenstein [36] are adopted in the present model for this reaction.

$$\alpha A_2CH_2 + C_2H_2 = bC_{13}H_{10} + H \quad (R14)$$

$$\alpha A_2CH_2 + C_2H_2 = A_2R_6 + H \quad (R15)$$

Two complementary pathways for phenalene formation are also included in the present model: $C_{10} + C_3$ and $C_{12} + C_1$ (Scheme 2b). Propargyl addition to naphth-1-yl is quite similar to the combination of two propargyl radicals (R16 and R17) [36]. The adduct intermediate either cyclizes to a benzene-like structure (phenalene, $A_2R_6$) or firstly cyclizes to a fulvene-like structure ($A_2R_5CH_2$) and then isomerizes to phenalene. The addition reactions of $C_3H_4$ isomers to naphth-1-yl radical are also included in the present model (R18 and R19). The rate constants of these reactions are analogized to the reactions of $C_3H_4$ isomers and propargyl [37,38]. Methyl additions to $C_{12}$ species, such as acenaphthylene, acenaphthyl and acenaphthenyl, kinetically connect to $A_2R_5CH_2$. The reaction scheme of $C_{12} + C_1$ is quite similar to $C_5 + C_1$ for benzene for-




**Scheme 2.** Reaction schemes of (a) $\alpha A_2CH_2 + C_2H_2$, (b) $A_2\text{-} + C_3H_3$ and $C_{12} + CH_3$, and the subsequent reaction steps for the formation of phenalene and benzo-indene. One- and two-way arrows indicate dehydrogenation and isomerization steps, respectively; double arrows indicate entrance channels.

mation; therefore, the rate constants of these reactions (R20 – R23) are analogized to the previous theoretical calculations of $C_5 + C_1$ reactions [39].

$$A_2\text{-} + C_3H_3 = A_2R_6 \quad \text{(R16)}$$

$$A_2\text{-} + C_3H_3 = A_2R_5CH_2 \quad \text{(R17)}$$

$$A_2\text{-} + aC_3H_4 = A_2R_6 + H \quad \text{(R18)}$$

$$A_2\text{-} + pC_3H_4 = A_2R_6 + H \quad \text{(R19)}$$

$$A_2R_5 + CH_3 = A_2R_5HCH_3\text{-} \quad \text{(R20)}$$

$$A_2R_5\text{-} + CH_3 = A_2R_5CH_2 \quad \text{(R21)}$$

$$A_2R_5H + CH_3 = A_2R_5HCH_3\text{-} + H \quad \text{(R22)}$$

$$A_2R_5CH_2 + H = A_2R_5HCH_3\text{-} \quad \text{(R23)}$$

### 3.2.3. Addition of $C_3$ species to naphth-1-yl-methyl radical

The reaction of propargyl and naphth-1-yl-methyl (R24) is proposed as a formation pathway of phenanthrene ($A_3$) in the present model. Similar to the reaction of propargyl and benzyl [40], the combination of propargyl and naphth-1-yl-methyl yields methyl-benzoindenyl radical ($bC_{13}H_8CH_3$). It then eliminates an H-atom to form methylene-benzoindene ($bC_{13}H_8CH_2$) and phenanthrene. Methylene-benzoindene is a fulvene-like intermediate and it isomerizes to phenanthrene, a peri–condensed aromatic structure. Meanwhile, the reactions of $bC_{13}H_8CH_3$ and $bC_{13}H_8CH_2$ connect to another phenanthrene formation pathway *via* methyl addition to benzoindenyl (R25 and R26). This reaction scheme from benzoindenyl to phenanthrene is again similar to the $C_5 + C_1$ pathway for benzene formation [39].

$$\alpha A_2CH_2 + C_3H_3 = bC_{13}H_8CH_3 + H \tag{R24}$$

$$bC_{13}H_9 + CH_3 = bC_{13}H_8CH_3 + H \tag{R25}$$

$$bC_{13}H_9 + CH_3 = bC_{13}H_8CH_2 + H_2 \tag{R26}$$

*3.2.4. Addition of $C_4$ species to naphth-1-yl-methyl radical*

Butadiyne addition to naphth-1-yl-methyl is quite similar to that of the acetylene addition reaction, shown in Scheme 2a. The first step of this reaction has two reaction channels, yields either ethynyl-benzo[*g*]indene or ethynyl-phenalene, which subsequently isomerize to 4*H*-cyclopenta[*cd*]phenalene. We propose one global reaction (R27) for this pathway using the rate constant of naphth-1-yl and acetylene reaction [41]. Acetylene or ethynyl addition to benzo[*g*]indenyl or phenalenyl radicals also produce $C_{15}H_{10}$ species (R28 – R32). In the present model, 4*H*-cyclopenta[*cd*]phenalene and 2a*H*-cyclopenta[*cd*]phenalene are lumped as one species ($cpC_{15}H_{10}$–2a) considering their similar formation pathways, but these are separated from 1*H*-cyclopenta[*cd*]phenalene because the latter is mainly derived from methyl-phenanthrene.

$$\alpha A_2CH_2 + C_4H_2 = cpC_{15}H_{10}\text{–}2a + H \tag{R27}$$

$$A_2R_6\text{-} + C_2H_2 = A_2R_6CHCH \tag{R28}$$

$$A_2R_6\text{-} + C_2H = A_2R_6C_2H \tag{R29}$$

$$cpC_{15}H_{10}\text{–}2a + H = A_2R_6CHCH \tag{R30}$$

$$A_2R_6C_2H + H = cpC_{15}H_{10}\text{–}2a + H \tag{R31}$$

$$bC_{13}H_9 + C_4H_2 = cpC_{15}H_{10}\text{–}2a + H \tag{R32}$$

*3.2.5. Addition of cyclopentadienyl and indenyl to naphth-1-yl-methyl radical*

The reaction of benzyl and indenyl was theoretically investigated by Sinha et al. [42] and was included in our previous indene model [17]. Due to the similarity between the molecular structures of benzyl and naphth-1-yl-methyl, the addition reactions of indenyl and cyclopentadienyl to naphth-1-yl-methyl are accordingly proposed in the present model (R33 – R40). Scheme 3a illustrates the reaction sequence after the combination of naphth-1-yl-methyl and indenyl. Intermediate $A_2CH_2C_9H_7$ loses an H-atom *via* C–H bond fission or H-abstraction to produce $A_2CHC_9H_7$ radical. Ring-expansion isomerization and following $\beta$-scission of C–H bond forms either 1,1-binaphthalene ($biA_2$) or 1,2-binaphthalene ($bi_{12}A_2$). Further dehydrogenation and cyclization of 1,1- and 1,2-binaphthalene forms perylene (PERY) and benzo-fluoranthene (BFTH), respectively. Benzo[*j*]fluoranthene and benzo[*k*]fluoranthene are both possible products of the latter reaction. In the present model, we only included the route for perylene but ignored the branching route for benzo-fluoranthene because the main objective of this study is the PAH chemistry of naphth-1-yl and naphth-1-yl-methyl. Our further study on the chemistry of naphth-2-yl would improve this part of PAH formation pathways. The reaction sequence between cyclopentadienyl and naphth-1-yl-methyl is exactly the same as that of indenyl and naphth-1-yl-methyl, illustrated in Scheme 3b. The promising products of cyclopentadienyl and naphth-1-yl-methyl reaction are fluoranthene, naphth-1-yl, and benzene. The rate constants of all reaction steps, (R33 – R40), are taken from the calculations of Sinha et al. [42].

$$C_9H_7 + \alpha A_2CH_2 = A_2CH_2C_9H_7 \tag{R33}$$

$$A_2CH_2C_9H_7 = A_2CHC_9H_7 + H \tag{R34}$$

$$A_2CHC_9H_7 = A_2A_2H \tag{R35}$$

$$biA_2 + H = A_2A_2H \tag{R36}$$

$$C_5H_5 + \alpha A_2CH_2 = A_2CH_2C_5H_5 \tag{R37}$$

$$A_2CH_2C_5H_5 = A_2CHC_5H_5 + H \tag{R38}$$

$$A_2CHC_5H_5 = C_{10}H_8C_6H_5 \tag{R39}$$

$$C_{10}H_7C_6H_5 + H = C_{10}H_8C_6H_5 \tag{R40}$$

*3.2.6. Addition of phenyl and naphth-1-yl to naphth-1-yl-methyl radical*

Addition reactions of phenyl and naphth-1-yl to naphth-1-yl-methyl are similar to each other (R41 – R47), as shown in Scheme 4. 1-Benzyl-naphthalene ($A_2CH_2A_1$) adduct loses an H-atom *via* C–H bond fission or H-abstraction to produce $A_2CHA_1$ radical. Subsequent H-shift, cyclization, and dehydrogenation steps convert $A_2CHA_1$ radical to 7*H*-benzo[*de*]anthracene (BdAN). Similarly, naphth-1-yl addition to naphth-1-yl-methyl forms 7*H*-benzo[*ij*]tetraphene (BiTEPH). Another entrance channel for 1-benzyl-naphthalene is the combination of naphth-1-yl and benzyl. The rate constants for the combination steps of these reactions (R41, R44, and R45) are taken from the suggested values of benzyl and phenyl reaction by Yuan et al. [12,13], and those for the cyclization and H-elimination reaction steps are analogized to the similar reactions of cyclohexene (R42, R43, R46, and R47) [43].

$$\alpha A_2CH_2 + A_2\text{-} = A_2CH_2A_2 \tag{R41}$$

$$biTEPH\text{–}H = A_2CHA_2 \tag{R42}$$

$$biTEPH\text{–}H = biTEPH + H \tag{R43}$$

$$\alpha A_2CH_2 + A_1\text{-} = A_2CH_2A_1 \tag{R44}$$

$$A_1CH_2 + A_2\text{-} = A_2CH_2A_1 \tag{R45}$$

$$BdANH = A_2CHA_1 \tag{R46}$$

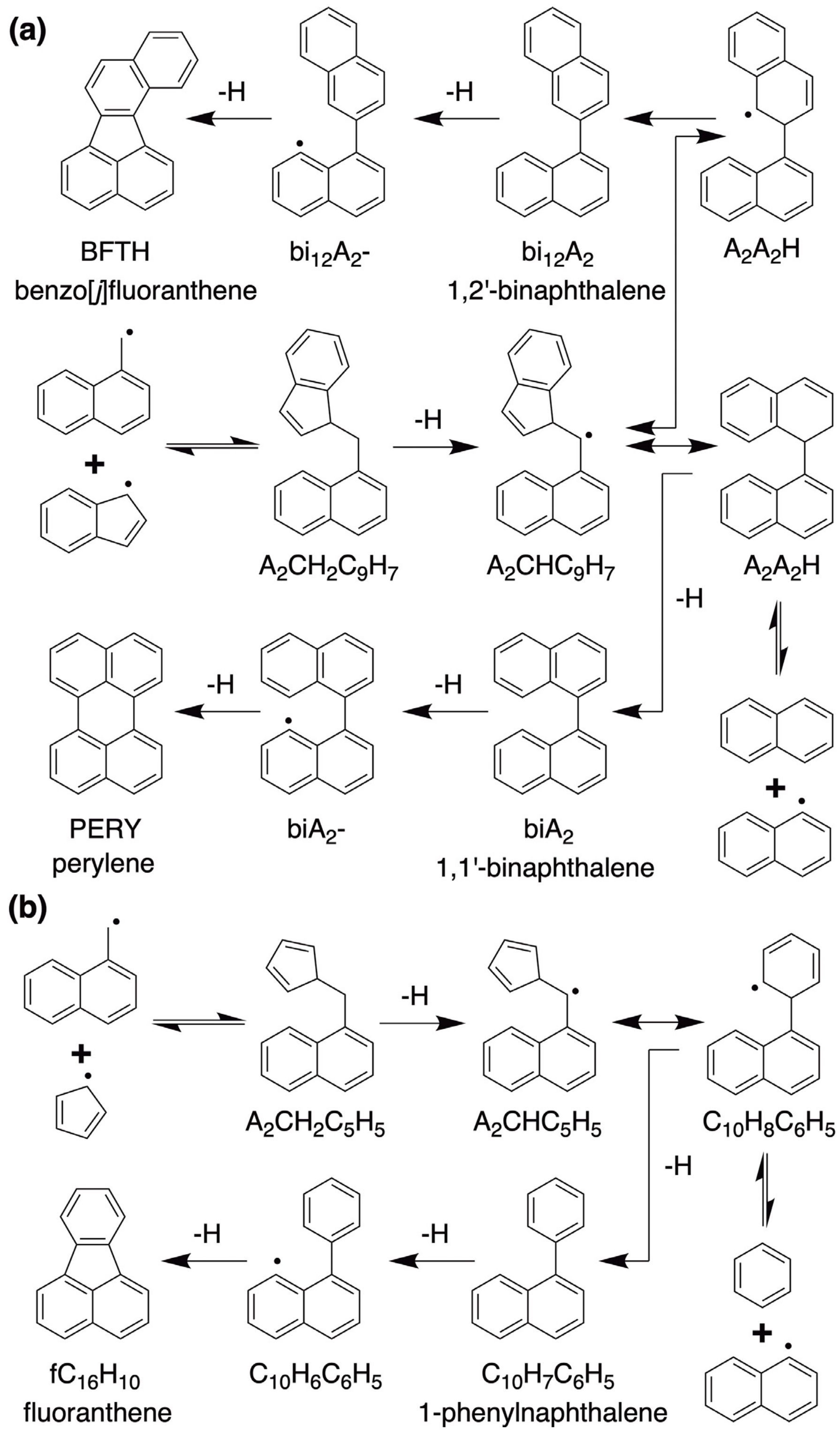
$$BdANH = BdAN + H \tag{R47}$$

*3.3. PAH pathways from methyl-naphthyl radical*

Of the two primary radicals formed from the decomposition of methyl-naphthalene, we consider methyl-naphthyl to be not as important as naphth-1-yl-methyl in PAH growth due to its shorter lifetime and lower concentration. Therefore, only limited numbers of PAH formation pathways are included in the present model for methyl-naphthyl. Generally, reactions of methyl-naphthyl are analogized to those of naphthyl and phenyl radicals.

*3.3.1. Methyl addition to methyl-naphthyl radical*

The addition of methyl to methyl-naphthyl forms dimethyl-naphthalene (R48). The reactions of dimethyl-naphthalene in the present model are analogized to those of toluene [12]. Its C–H bond fission (R49) and H-abstraction reactions produce $CH_3A_2CH_2$ radical. Methyl addition to $\alpha$-methyl-naphthalene also contributes to dimethyl-naphthalene formation (R50). Additionally, ipso-reaction between methyl-naphthalene and methyl-naphthyl radical (R51) leads to the formation of dimethyl-naphthalene. Decomposition of $CH_3A_2CH_2$ radical may form xylylene structure ($CH_2A_2CH_2$) or benzo-fulvenallene (R52 and R53). $CH_2A_2CH_2$ may




**Scheme 3.** Reaction schemes of (a) $\alpha A_2CH_2 + C_9H_7$ and (b) $A_2CH_2 + C_5H_5$, and the subsequent reaction steps for benzo-fluoranthene and fluoranthene formation. One- and two-way arrows indicate dehydrogenation and isomerization steps, respectively; double arrows indicate entrance channels.

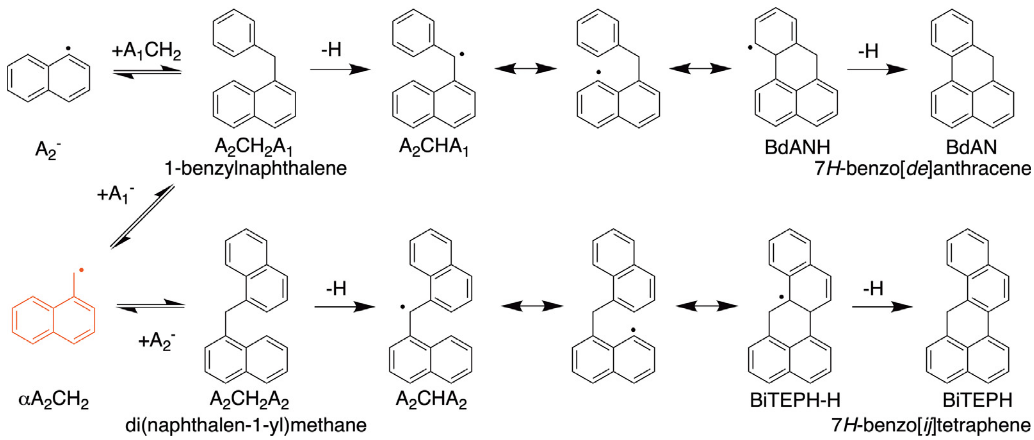
isomerize to vinyl-naphthalene (R54), as proposed previously for 5,6-dimethylenecyclohexa-1,3-diene [44]. The rate constants of R52 – R54 are referred to the reactions of xylylene [44]. H-addition followed by $CH_3$-elimination reactions convert $CH_3A_2CH_2$ radical back to naphth-1-yl-methyl and methyl-naphthyl radicals (R55 and R56).

$$CH_3A_2CH_3\ (+M) = C_{10}H_6CH_3 + CH_3\ (+M) \tag{R48}$$

$$CH_3A_2CH_3\ (+M) = CH_3A_2CH_2 + H\ (+M) \tag{R49}$$

$$CH_3A_2CH_3 + H = \alpha A_2CH_3 + CH_3 \tag{R50}$$

$$\alpha A_2CH_3 + C_{10}H_6CH_3 = CH_3A_2CH_3 + A_2\text{-} \tag{R51}$$




**Scheme 4.** Reaction schemes of $\alpha A_2CH_2$ + $A_2$-, $\alpha A_2CH_2$ + $A_1$-, and $A_2$- + $A_1CH_2$, as well as the subsequent reaction steps for the formation of 7*H*-benzo[*ij*]tetraphene and 7*H*-benzo[*de*]anthracene. One- and two-way arrows indicate dehydrogenation and isomerization steps, respectively; double arrows indicate entrance channels.

$$CH_3A_2CH_2 = CH_2A_2CH_2 + H \quad (R52)$$

$$CH_3A_2CH_2 = C_9H_6CCH_2 + CH_3 \quad (R53)$$

$$CH_2A_2CH_2 = A_2C_2H_3 \quad (R54)$$

$$CH_3A_2CH_2 + H = \alpha A_2CH_2 + CH_3 \quad (R55)$$

$$CH_3A_2CH_2 + H = C_{10}H_6CH_3 + CH_3 \quad (R56)$$

We consider the possibility that two methyl groups may exist at both sides of the *zigzag* periphery of naphthalene (sites 1 and 8); dehydrogenation and cyclization of this intermediate leads to acenaphthene ($A_2R_5H_2$). This reaction (R57) is proposed in the present model by referring to the cyclization reaction of naphth-1-yl-vinyl radical [14].

$$CH_3A_2CH_2 = A_2R_5H_2 + H \quad (R57)$$

### 3.3.2. Addition of $C_2$ species to methyl-naphthyl radical

Since methyl-naphthyl radical is a set of methyl-naphthyl isomers with their radical sites from 2 to 8, acetylene addition to methyl-naphthyl may occur at any radical site. Only the reaction on site 8 is included in the present model because its further cyclization forms phenalene. The rate constant of the addition step is assumed to the same as phenyl + acetylene reaction [41], and the rate constants of cyclization and dehydrogenation reactions are analogized to the chemistry of cyclohexene [43].

### 3.3.3. Addition of $C_4$ species to methyl-naphthyl radical

H-abstraction vinylacetylene-addition (HAVA) mechanism is adopted for methyl-naphthyl radical, yielding 1-/9-methyl-phenanthrene ($MPA_3$–1/–9). The rate constant of HAVA mechanism is taken from the recommendation of Blanquart et al. [14]. For naphthalene formation, Richter and Howard [45] suggested the combination of phenyl and but–1-en-3-yn-1-yl radicals, while Li et al. [46] proposed the reaction between benzene and but–1-en-3-yn-1-yl radical as an alternate pathway. We considered similar reactions for methyl-naphthyl in the present model.

### 3.3.4. Addition of $C_6$ and $C_{10}$ species to methyl-naphthyl radical

The addition reactions of phenyl and naphth-1-yl to methyl-naphthyl are quite similar, and their rate constants are analogized to the combination reaction of two phenyl radicals [27]. Methyl-phenyl-naphthalene ($A_1A_2CH_3$) is the product of the reaction between phenyl and methyl-naphthyl. Methyl-phenyl-naphthalene is a lumped species representing this class of isomers. Its thermodynamic and transport data are evaluated by the structure of 1-methyl-5-phenylnaphthalene. Further reaction of methyl-phenyl-naphthalene form methyl-fluoranthene ($C_{16}H_9fCH_3$) *via* stepwise dehydrogenation. Methyl-binaphthalene ($biA_2CH_3$) is the product of the reaction between naphth-1-yl and methyl-naphthyl. Methyl-binaphthalene is not a specific molecule but a lumped species representing a class of methyl-binaphthalene isomers. In the present model, its thermodynamic and transport data are evaluated by the structure of 5-methyl-1,1-binaphthalene. Methyl-binaphthalene cyclizes to methyl-perylene ($PERYCH_3$) *via* stepwise dehydrogenation. Other aromatization reactions of methyl-binaphthalene are not included in the present model, which form methyl-benzo-fluoranthene, benzo-tetraphene, etc. Addition-elimination reactions are also considered here, such as methyl-naphthalene addition to phenyl and naphth-1-yl, forming methyl-phenyl-naphthalene and methyl-binaphthalene, respectively.

## 3.4. PAH pathways from benzo-fulvenallenyl radical

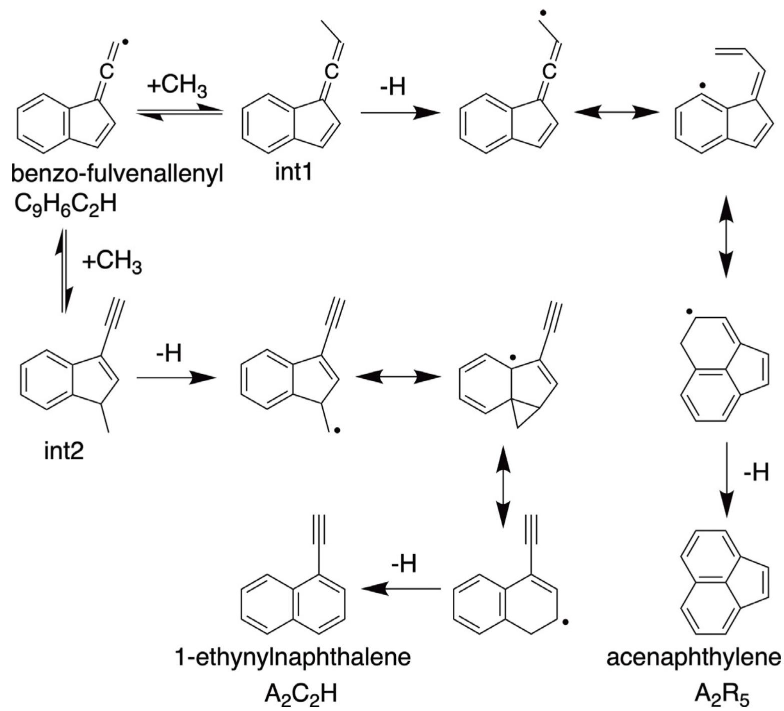
The kinetic role of benzo-fulvenallenyl ($C_9H_6C_2H$) in the chemistry of naphth-1-yl-methyl is almost the same as that of fulvenallenyl for benzyl. However, due to the limited investigation on the chemistry of fulvenallenyl radical [20,21], the reaction network of benzo-fulvenallenyl is quite challenging to be comprehensively described in the present model. We propose a series of mass growing reaction pathways from benzo-fulvenallenyl, as introduced in this section, which may sensitively impact the PAH formation chemistry.

### 3.4.1. Addition of small hydrocarbons to benzo-fulvenallenyl radical

Since benzo-fulvenallenyl is an RSR, the addition reactions may occur at different positions. For example, methyl addition may attach to its allene group or its penta-ring (int1 or int2 in Scheme 5); subsequent dehydrogenation and cyclization steps convert int1 and int2 to acenaphthylene and 1-ethynyl-naphthalene, respectively. We propose two global reactions (R58 and R59) for the pathways shown in Schemes 5. Reaction R59 is analogized to $C_1$ + $C_5$ pathway for benzene formation, and the rate constant is taken from the theoretical calculation of Sharma and Green [39]. The rate constant of R58 is assumed to be equivalent to R59.

$$C_9H_6C_2H + CH_3 = A_2R_5 + 2H \quad (R58)$$

$$C_9H_6C_2H + CH_3 = A_2C_2H + 2H \quad (R59)$$




**Scheme 5.** Reaction scheme of $C_9H_6C_2H$ + $CH_3$ and its subsequent reaction steps forming acenaphthylene and 1-ethynyl-naphthalene. One- and two-way arrows indicate dehydrogenation and isomerization steps, respectively; double arrows indicate entrance channels.

The addition of acetylene to benzo-fulvenallenyl is a possible formation pathway for fluorenyl radical. This reaction is similar to the combination of acetylene and but–1-en-3-yn-1-yl for phenyl formation, which is, however, deemed unlikely according to previous studies [47,48]. Therefore, the reaction of acetylene and benzo-fulvenallenyl is not proposed in the present model. Instead, we propose the reaction between acetylene and vinyl-indenyl for fluorene formation (R60), similar to the reaction of acetylene and vinyl-cyclopentadienyl for indene formation [49].

$$C_9H_6C_2H_3 + C_2H_2 = C_{13}H_{10} + H \tag{R60}$$

Previous models have proposed the formation of two- / three-ring aromatic hydrocarbons *via* the reactions of fulvenallenyl [21,50]. Similar to the naphthalene formation through the reaction of propargyl and fulvenallenyl [20], the reaction of propargyl and benzo-fulvenallenyl is proposed to form phenanthrene (R61). The rate coefficient of this reaction is analogized to the self-combination reaction of propargyl radical, as suggested for propargyl and fulvenallenyl in literature [50]. Matsugi and Miyoshi [21] proposed to consider steric hindrance by a factor of $(T/300)^{0.3}$ for fulvenallenyl + propargyl while adopting the rate constant of propargyl self-combination reaction. We estimate that this steric hindrance effect for benzo-fulvenallenyl may reduce the rate coefficient by a factor of 15. According to benzene formation *via* the reaction between $C_3H_4$ isomers and propargyl [37,38], we propose phenanthrene formation pathways *via* the addition of propargyl to benzo-fulvenallene or ethynyl-indene (R62 and R63).

$$C_9H_6C_2H + C_3H_3 = A_3 \tag{R61}$$

$$C_9H_6CCH_2 + C_3H_3 = A_3 + H \tag{R62}$$

$$C_9H_7C_2H + C_3H_3 = A_3 + H \tag{R63}$$

*3.4.2. Addition of aromatics to benzo-fulvenallenyl radical*

Self-addition of fulvenallenyl has been proposed in literature to form phenanthrene [20], thus self-addition of benzo-fulvenallenyl is proposed here to form picene (R64). We adopted the rate constants of propargyl + propargyl for this reaction, though we reduced the rate constant by a factor of 200 considering the steric hindrance of two benzo-fulvenallenyl radicals ($15^2$). The addition reactions of benzo-fulvenallene or ethynyl-indene to benzo-fulvenallenyl are also included in the present model (R65 and R66). Another similar PAH formation pathway is the reaction between benzo-fulvenallenyl and fulvenallenyl (R67) to form benzo[*ghi*]fluoranthene (BgF).

$$2C_9H_6C_2H = \text{picene} \tag{R64}$$

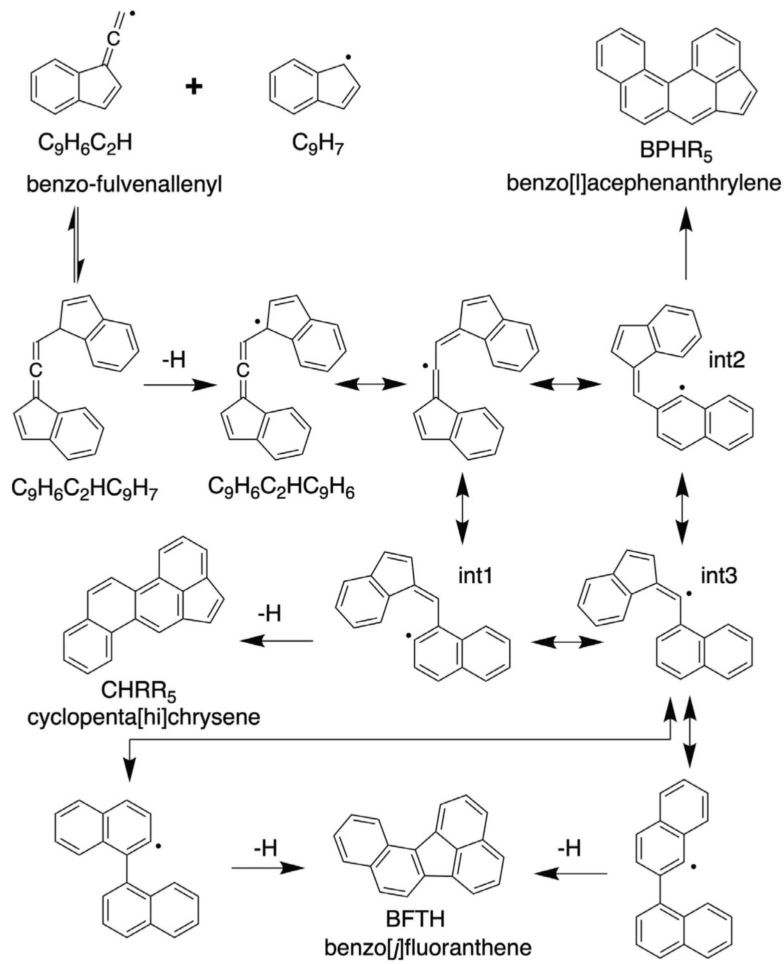
$$C_9H_6CCH_2 + C_9H_6C_2H = \text{picene} + H \tag{R65}$$

$$C_9H_7C_2H + C_9H_6C_2H = \text{picene} + H \tag{R66}$$

$$C_9H_6C_2H + C_7H_5 = \text{BgF} + 2H \tag{R67}$$

Indenyl is an important RSR intermediate in the pyrolysis of $\alpha$-methyl-naphthalene. Its reaction with benzo-fulvenallenyl may contribute to $C_{20}$ PAH formation. The reaction of fulvenallenyl and cyclopentadienyl has been shown by Da Silva and Bozzelli [20] to form biphenyl *via* ring expansion rearrangement after their combination. A similar reaction between benzo-fulvenallenyl and indenyl is proposed in Scheme 6. The adduct formed from this reaction is $C_9H_6C_2HC_9H_7$, and its H-elimination then forms $C_9H_6C_2HC_9H_6$ radical. Ring expansion reaction of $C_9H_6C_2HC_9H_6$ may yield two intermediates, int1 and int2 in Scheme 6. The following cyclization and dehydrogenation of int1 and int2 produce benzo[*l*]acephenanthrylene ($BPHR_5$) and cyclopenta[*hi*]chrysene ($CHRR_5$), respectively. H-shift isomerization of int1 and int2 forms int3, which may undergo the 2nd stage of ring expansion to form benzo[*j*]fluoranthene (BFTH). A global reaction (R68) is proposed for this reaction in the present model using the rate constant of cyclopentadienyl recombination reaction [14].

$$C_9H_6C_2H + C_9H_7 = 0.2\text{BFTH} + 0.4\text{BPHR}_5 + 0.4\text{CHRR}_5 + 2\text{H} \quad \text{R68}$$




**Scheme 6.** Reaction schemes of $C_9H_6C_2H$ + $C_9H_7$, and subsequent reaction steps for the formation of benzo[*l*]acephenanthrylene, cyclopenta[*hi*]chrysene, and benzo[*j*]fluoranthene. One- and two-way arrows indicate dehydrogenation and isomerization steps, respectively; double arrows indicate entrance channels.

The reaction of naphth-1-yl and benzo-fulvenallenyl is proposed in the present model (R69) to form 7*H*-benzo[*ij*]tetraphene, as shown in Scheme 7. The adduct from the addition step undergoes ring expansion of the benzo-fulvenallene structure, then cyclize a new ring containing naphthalene part. Undergoing a similar reaction sequence, the combination of benzo-fulvenallenyl and naphth-1-yl-methyl may produce dihydropicene ($PC_{22}H_{16}$, R72). Its further dehydrogenation again yields picene. The addition reactions of benzo-fulvenallene or ethynyl-indene to naphth-1-yl or naphth-1-yl-methyl also form 7*H*-benzo[*ij*]tetraphene and dihydropicene (R70, R71, R73, and R74), respectively. The rate constants of these reactions are taken from the reactions of fulvenallene and fulvenallenyl in previous models [12,17,50].

$$C_9H_6C_2H + A_2\text{-} = \mathrm{BiTEPH} \tag{R69}$$

$$C_9H_6CCH_2 + A_2\text{-} = \mathrm{BiTEPH} + H \tag{R70}$$

$$C_9H_7C_2H + A_2\text{-} = \mathrm{BiTEPH} + H \tag{R71}$$

$$C_9H_6C_2H + \alpha A_2CH_2 = PC_{22}H_{16} \tag{R72}$$

$$C_9H_6CCH_2 + \alpha A_2CH_2 = PC_{22}H_{16} + H \tag{R73}$$

$$C_9H_7C_2H + \alpha A_2CH_2 = PC_{22}H_{16} + H \tag{R74}$$

### *3.5. Thermodynamic and transport data*

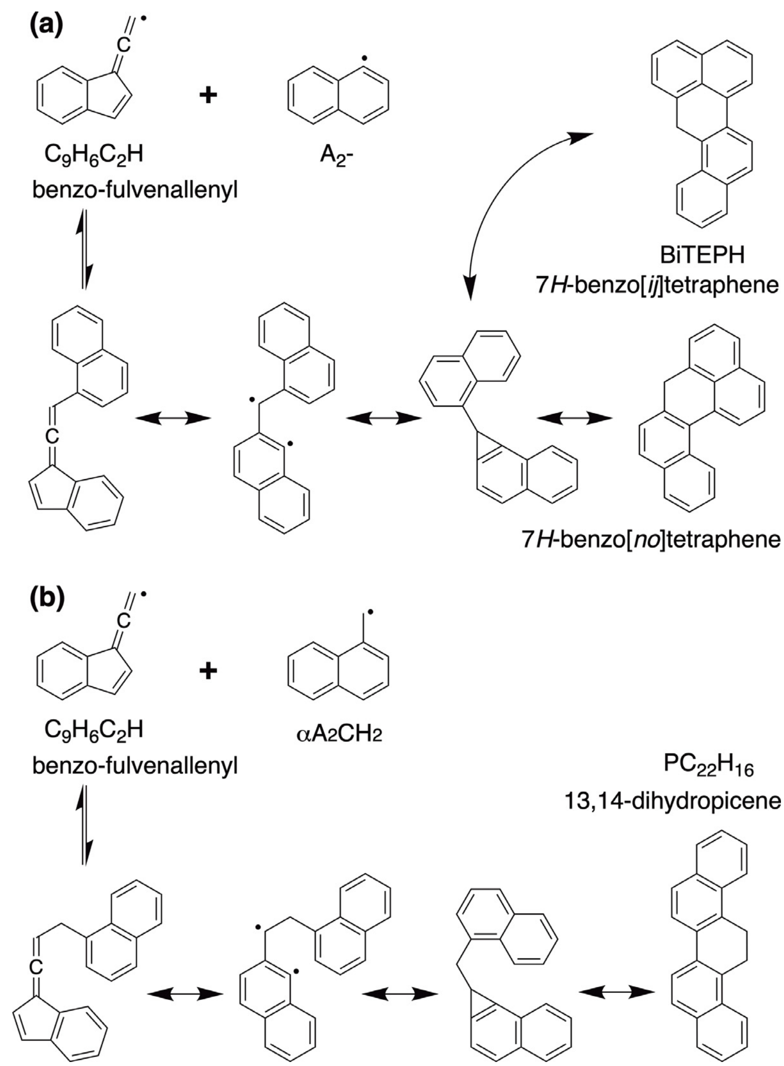
Thermodynamic and transport data of the species in the present model are mainly taken from our indene model [17]. Values for the newly added species are calculated by the group-additivity method [51,52].

## 4. Results and discussion

### *4.1. Experimental results and model predictions*

In this section, we present the experimental and numerical results of PAHs formed from $\alpha$-methyl-naphthalene pyrolysis in a flow reactor at 30 and 760 Torr. The major molecular structures of the aromatic products are speculated according to literature recommendations. Mole fractions of the identified species are evaluated according to the method introduced in our previous study [17] (and SMM2). Numerical simulations of the flow reactor are performed using Cantera [53] with temperature profiles measured in the experiment used as input parameters (SMM1). Dozens of aromatic products were detected in the experiment, and their mole fraction profiles are plotted as a function of reaction temperature and compared with simulated profiles in Figs. 1-5. In general, the present model not only adequately captures the formation temperature windows of the PAHs but also reasonably captures their absolute values along reaction temperature.

The number of cyclic rings ($N_c$) in a PAH indicates its degree of aromatization. The bigger the $N_c$, the higher C/H ratio the molecule




**Scheme 7.** Reaction schemes of (a) $C_9H_6C_2H$ + $A_2$- and (b) $C_9H_6C_2H$ + $\alpha A_2CH_2$, and the subsequent reaction steps for the formation of 7*H*-benzo[*ij*]tetraphene and dihydropicene. Two-way arrows indicate isomerization steps, double arrows indicate entrance channels.

has, and the more dehydrogenation steps this molecule needed to undergo during its formation process. The possible molecular structures of each mass peak, measured during $\alpha$-methyl-naphthalene pyrolysis, are discussed in this section by considering $N_c$ and the kinetics of their formation and further PAH growth.

#### 4.1.1. Bicyclic and tricyclic aromatics

The molecular structure of $C_{12}H_{12}$ is assumed to be dimethyl-naphthalene or ethyl-naphthalene [8,10]. The decomposition of $\alpha$-methyl-naphthalene produces methyl-naphthyl, naphth-1-yl-methyl and methyl radicals [10]. The combination of methyl radical with two $C_{11}H_9$ radicals yields $C_{12}H_{12}$ isomers. Fig. 1d plots mole fractions of $C_{12}H_{12}$ isomers which are calibrated as dimethyl-naphthalene. The present model shows higher concentrations of dimethyl-naphthalene than ethyl-naphthalene at both pressures. In particular, the formation of ethyl-naphthalene is negligible at 30 Torr. C–C bond energy between the benzene ring and methyl group is stronger than that in an alkane carbon chain. Thus dimethyl-naphthalene shows higher thermal stability than ethyl-naphthalene at pyrolytic conditions. The reaction between methyl and naphth-1-yl-methyl consumes ethyl-naphthalene, while methyl reaction with methyl-naphthyl has a positive contribution to dimethyl-naphthalene formation. The major formation pathways for both dimethyl-naphthalene and ethyl-naphthalene are ipso-reactions: methyl-naphthyl and naphth-1-yl-methyl radicals taking methyl groups from $\alpha$-methyl-naphthalene. As a result, the present model predicts the formation of $C_{12}H_{12}$ isomers at relatively low temperatures.

$C_{12}H_{10}$ in Fig. 1c is considered as the dehydrogenation product of $C_{12}H_{12}$. Three candidates molecular structures of $C_{12}H_{10}$ in literature [34,54-56] are acenaphthene, vinyl-naphthalene, and biphenyl. We evaluated the mole fraction of $C_{12}H_{10}$ by the estimated PICS of acenaphthene. Vinyl-naphthalene can hardly be formed from the H-elimination of ethyl-naphthalene. Modeling results reveal that the reaction of naphth-1-yl-methyl and methyl yields $A_2CHCH_3$ radical as an intermediate for vinyl-naphthalene formation. Biphenyl is also not that easy to be formed *via* benzyne ($oC_6H_4$) → phenyl → biphenyl, although benzyne has a large concentration in $\alpha$-methyl-naphthalene pyrolysis. Fig. 1a presents the predictions of biphenylene from the combination of two benzyne molecules. The subsequent conversion from biphenylene to biphenyl is not very efficient, thus it cannot reproduce the mole fraction of $C_{12}H_{10}$. Previous models [34,57] proposed that acenaphthene is formed *via* the reaction of naphthalene and vinyl radical, and the isomerization of vinyl-naphthalene. This reaction pathway also cannot reproduce the experimental measurements, as indicated by the predicted mole fraction of vinyl-naphthalene in Fig. 1c (dash line). The cyclization and H-elimination reaction of

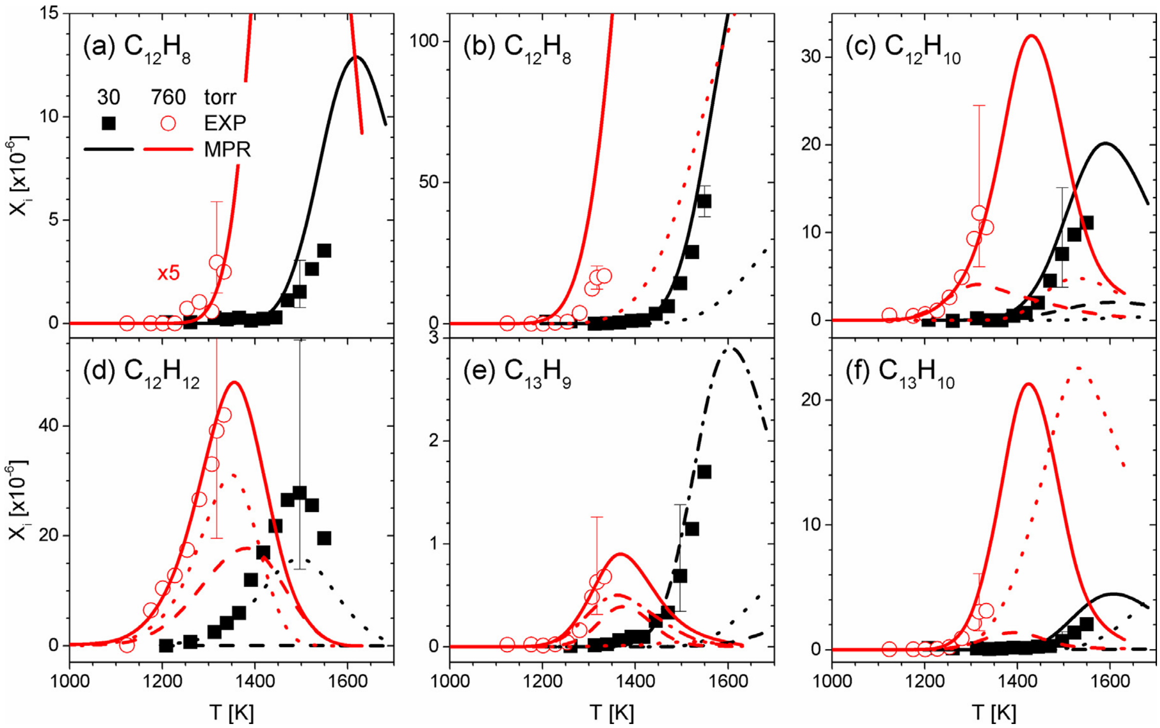



**Fig. 1.** Mole fraction profiles of $C_{12}$ and $C_{13}$ PAHs measured in the experiment (scatter, EXP) and simulated by the present model (lines, MPR). (a) $C_{12}H_8$: biphenylene (measurements and simulations are multiplied by a factor of 5 at 760 Torr); (b) $C_{12}H_8$: solid line for acenaphthylene, dot line for 1-ethynyl-naphthalene; (c) $C_{12}H_{10}$: solid line for acenaphthene, dash line for 1-vinyl-naphthalene, dot line for biphenyl; (d) $C_{12}H_{12}$: dash line for 1-ethyl-naphthalene, dot line for dimethyl-naphthalene, solid line for their sum; (e) $C_{13}H_9$: dash line for benzo-indenyl, dot line for fluorenyl, dash-dot line for phenalenyl, solid line for the their sum; (f) $C_{13}H_{10}$: dash line for benzo-indene, dot line for fluorene, solid line for phenalene.

$CH_3A_2CH_2$ radical, proposed in the present model, significantly improves model prediction. It contributes 96% and 38% of acenaphthene production at 30 and 760 Torr, respectively. Acenaphthene thus may be the major $C_{12}H_{10}$ isomer derived from dimethyl-naphthalene.

According to the modeling results in Fig. 1b, the major $C_{12}H_8$ component is acenaphthylene. Dehydrogenation of acenaphthene has a major contribution (63%) at 760 Torr, while the combination of benzo-fulvenallenyl and methyl is dominant (99%) at 30 Torr. The reaction of naphth-1-yl and acetylene is the main formation reaction of ethynyl-naphthalene. As acetylene is a highly unsaturated intermediate derived from methyl radical, ethynyl-naphthalene is thus formed at a higher temperature than acenaphthylene.

Phenalene, fluorene, and benzo-indene are three major $C_{13}H_{10}$ aromatics that may be formed in $\alpha$-methyl-naphthalene pyrolysis. The addition of acetylene and ethynyl to naphth-1-yl-methyl yields phenalene and benzo-indene, while the addition of acetylene to vinyl-indenyl yields fluorene [49]. Compared with naphth-1-yl-methyl reactions, those of propargyl and naphth-1-yl have relatively smaller contribution to $C_{13}H_{10}$ under both pressures. Fig. 1f presents model predictions of these species. Phenalene and fluorene have comparable mole fractions, while benzo-indene has a much smaller amount. H-loss of $C_{13}H_{10}$ isomers forms $C_{13}H_9$ radicals. The mole fraction distributions of three $C_{13}H_9$ radicals are similar than those of $C_{13}H_{10}$ isomers, as shown in Fig. 1e. Phenalenyl radical obviously has the highest amount.

Common $C_{14}H_{10}$ PAHs discussed in the literature are phenanthrene and anthracene [56]. Here, $C_{14}H_{10}$ is assumed to be phenanthrene considering the molecular structure of $\alpha$-methyl-naphthalene. Three main pathways proposed for phenanthrene in the present model (see Section 3) are: a) benzo-fulvenallenyl and propargyl route, b) naphth-1-yl-methyl and propargyl route, and c) fulvenallenyl combination route. Rate-of-production (ROP) analysis indicates that benzo-fulvenallenyl and propargyl combination route plays the major role (78% and 71% at 30 and 760 Torr, respectively), while the contribution of naphth-1-yl-methyl + propargyl route increases with pressure (<1% at 30 Torr and 11% at 760 Torr).

#### 4.1.2. Tetracyclic and pentacyclic aromatics

Cyclopenta[*def*]phenanthrene is a $C_{15}H_{10}$ molecular structure commonly proposed in previous studies [17,58]. Shukla et al. [59] recommended that methyl addition can take place at the armchair site of phenanthrene, which may be easier than acetylene or vinyl addition due to the steric crowding. Cyclization in the following step fuses cyclopenta[*def*]phenanthrene. The mole fraction of phenanthrene is at ppm level, as shown in Fig. 2a; its reaction with methyl, therefore, has negligible contribution to cyclopenta[*def*]phenanthrene. On the contrary, the reaction of naphth-1-yl-methyl and butadiyne has the advantage from the context of reactant concentrations to yield $C_{15}H_{10}$ aromatics. In the present model, we considered 2a*H*-cyclopenta[*cd*]phenalene ($cpC_{15}H_{10}$–2a) to be the product of this reaction, which is an isomer of cyclopenta[*def*]phenanthrene. Similar to the formation of acenaphthylene from the reaction of acetylene and naphth-1-yl, acetylene addition to phenalenyl may cross over the *zigzag* periphery to yield 2a*H*-cyclopenta[*cd*]phenalene. But this pathway is much less important than the direct reaction from naphth-1-yl-methyl (~55% at both pressures) in $\alpha$-methyl-naphthalene pyrolysis. Fig. 2b shows good predictions of the experimental data by these kinetic assumptions.

Literature study has identified four major $C_{16}H_{10}$ PAHs: fluoranthene, pyrene, acephenanthrylene, and aceanthrylene [58]. In the present model, we only considered fluoranthene and pyrene, as shown in Fig. 2c. $C_2$-addition to phenanthrene is proposed to form pyrene. The reaction flux of acephenanthrylene and aceanthrylene, that may also be derived from $C_2$-addition to phenanthrene and anthracene, is lumped to pyrene. Modeling results show that flu-

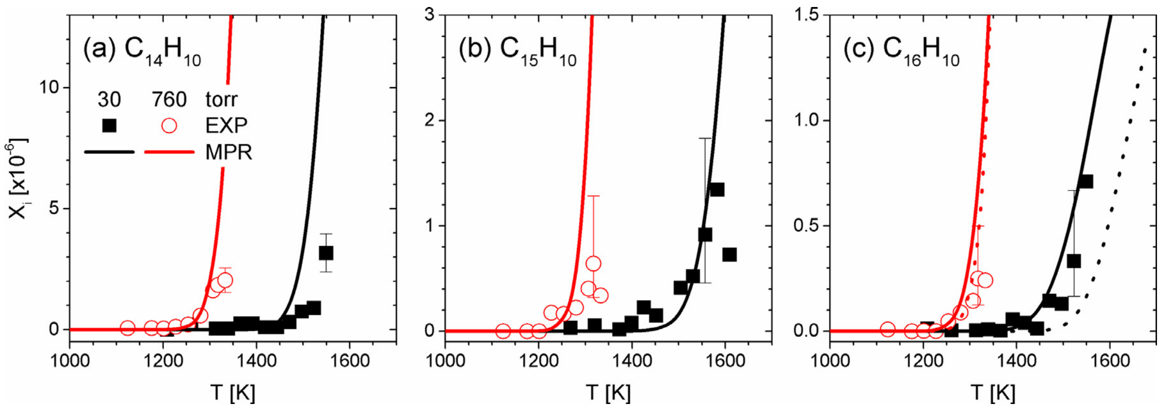



**Fig. 2.** Mole fraction profiles of $C_{14}$ - $C_{16}$ PAHs measured in the experiment (scatter, EXP) and simulated by the present model (lines, MPR). (a) $C_{14}H_{10}$: phenanthrene; (b) $C_{15}H_{10}$: 2a*H*-cyclopenta[*cd*]phenalene; (c) $C_{16}H_{10}$: solid line for fluoranthene, dot line for pyrene.

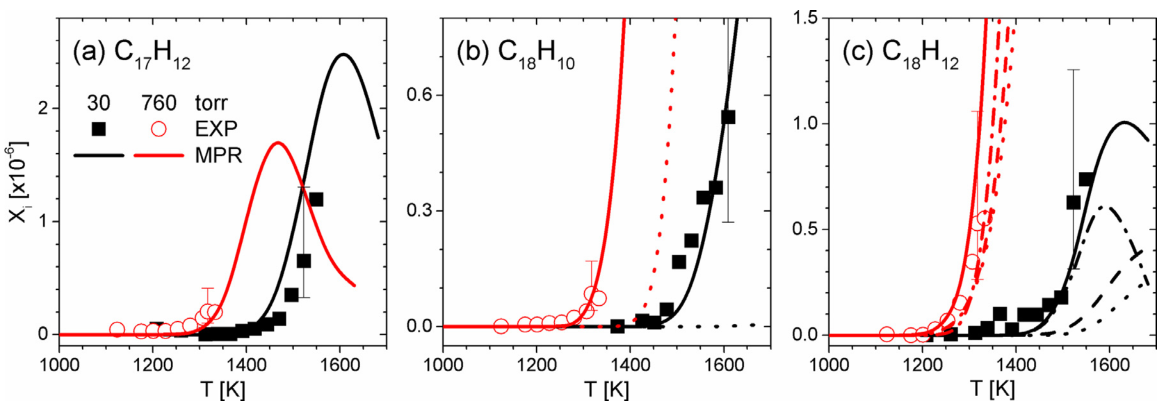



**Fig. 3.** Mole fraction profiles of $C_{17}$ and $C_{18}$ PAHs measured in the experiment (scatter, EXP) and simulated by the present model (lines, MPR). (a) $C_{17}H_{12}$: methyl-fluoranthene; (b) $C_{18}H_{10}$: solid line for benzo[*ghi*]fluoranthene, dot line for ethynyl-pyrene; (c) $C_{18}H_{12}$: dash line for benzo[*c*]phenanthrene, dot line for chrysene, dash-dot line for triphenylene, solid line for their sum.

oranthene has higher formation efficiency than pyrene at 30 Torr but it is almost equivalent at 760 Torr. Fluoranthene may be a decomposition production of methyl-fluoranthene; however, this is not the case in $\alpha$-methyl-naphthalene pyrolysis due to the large amount of benzyne. Naphthyl-phenyl, which is derived from the combination of naphth-1-yl and benzyne, cyclodehydrogenizes to fluoranthene. Sinha et al. [42] suggested that naphthyl-phenyl may also convert to pyrene, but modeling results find it a minor pathway. Methyl addition to 2a*H*-cyclopenta[*cd*]phenalene and subsequent ring expansion wins the competition for pyrene formation (39% and 77% at 30 and 760 Torr, respectively) because of the large amount of methyl in $\alpha$-methyl-naphthalene pyrolysis. This prototypical reaction is validated to be a considerable conversion route between penta- and benzene-ring PAHs, such as the conversion between indene and naphthalene [17,60]. The reaction between propargyl and phenalenyl radicals also fuses pyrene, which provides a contribution of ~ 10 – 30%.

The molecular structure of $C_{17}H_{12}$ aromatic hydrocarbon has not been discussed in detail in previous studies [17,61-64]. In the modeling study of indene pyrolysis, Jin et al. [17] proposed it to be benzo-fluorene, formed by the reaction between indene and indenyl. In the current study, the dominant $C_{17}H_{12}$ isomer is assumed to be methyl-fluoranthene because it has a direct formation reaction from methyl-naphthyl radical. Similar to fluoranthene, this reaction sequence includes benzyne addition to methyl-naphthyl, cyclization and dehydrogenation. Good agreement between experimental measurements and modeling results in Fig. 3a confirms the effectiveness of this kinetic assumption.

Fig. 3c presents the mole fractions of $C_{18}H_{12}$ PAHs. Chrysene, benzo[*c*]phenanthrene, and triphenylene are the three major $C_{18}H_{12}$ isomers included in the present model. In the previous indene model [17], clustering of hydrocarbons by radical chain reactions [65,66] is proposed as a major formation route for chrysene and benzo[*c*]phenanthrene, such as the reactions of cyclopentadienyl, indenyl, and benzo-indenyl radicals. Traditional H-abstraction acetylene-/vinylacetylene-addition routes [41,67] are also included for all the three $C_{18}H_{12}$ isomers besides the reactions of RSRs. We considered the self-addition route of benzyne, due to its large amount, for triphenylene formation: o-benzyne $\overset{1st}{\Rightarrow}$ biphenylene $\overset{2nd}{\Rightarrow}$ triphenylene [68]. Modeling results shown in Fig. 3c indicate significant efficiency of this reaction route at low pressure.

As shown in Fig. 3b, benzo[*ghi*]fluoranthene is the dominant component of $C_{18}H_{10}$, while ethynyl-pyrene has negligible concentration, particularly at 30 Torr. Traditional formation routes, such as the dehydrogenation of benzo[*c*]phenanthrene and acetylene addition to fluoranthene, cannot reproduce the measured mole fraction of $C_{18}H_{10}$. In the present model, two new pathways are proposed: a) the reaction of benzo-fulvenallenyl and fulvenallenyl (see Section 3.4.2), and b) the reaction of naphthylene ($A_2T_1$) and ethynyl-phenyl ($C_6H_4C_2H$). The first route is a typical RSR pathway; kinetic analysis of modeling results indicates that it contributes 78% and 94% to the formation of benzo[*ghi*]fluoranthene at 30 and 760 Torr, respectively.

Very few literature studies have discussed the possible molecular structures of $C_{19}H_{14}$ PAHs [69]. Methyl radical addition to

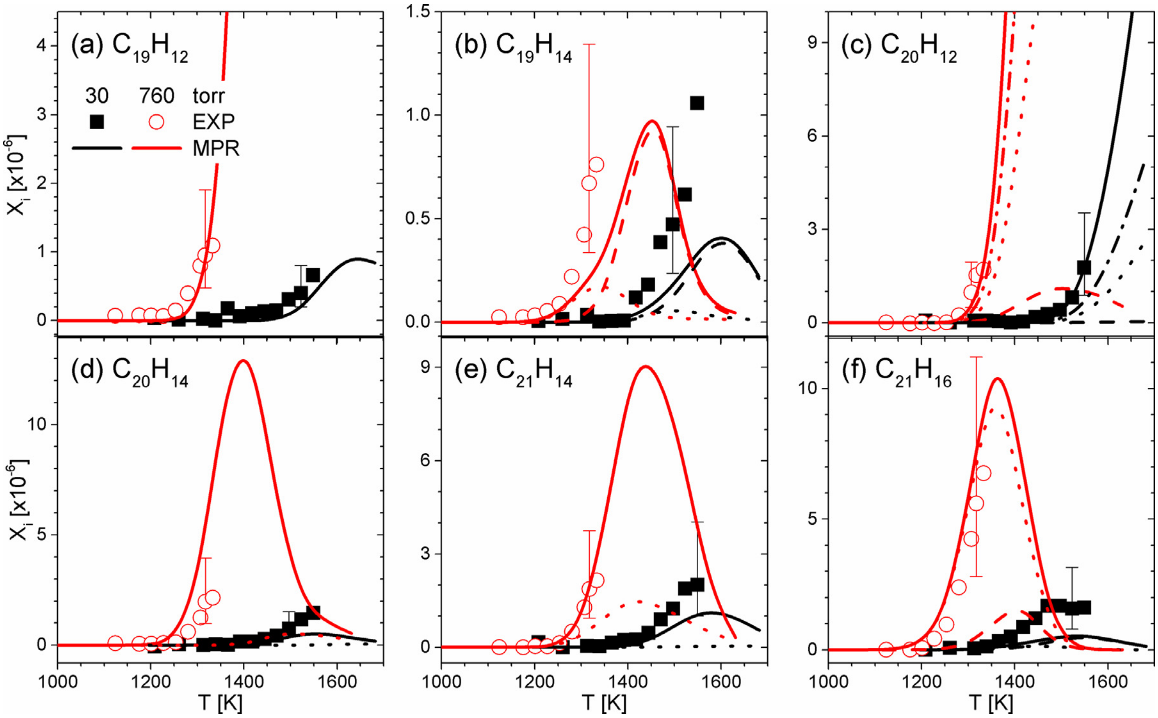



**Fig. 4.** Mole fraction profiles of $C_{19}$ – $C_{21}$ PAHs measured in the experiment (scatter, EXP) and simulated by the present model (lines, MPR). (a) $C_{19}H_{12}$: 4*H*-benzo[*cd*]fluoranthene; (b) $C_{19}H_{14}$: dot line for 3-phenyl-phenalene, dash line for methyl-triphenylene, solid line for their sum; (c) $C_{20}H_{12}$: dot line for benzo[*j*]fluoranthene, dash line for perylene, dash-dot line for benzo[*l*]acephenanthrylene and cyclopenta[*hi*]chrysene, sold line for their sum; (d) $C_{20}H_{14}$: dot line for phenyl-phenanthrene, solid line for binaphthalene; (e) $C_{21}H_{14}$: dot line for methyl-perylene, solid line for 7*H*-benzo[*ij*]tetraphene; (f) $C_{21}H_{16}$: dash line for methyl-binaphthalene, dash-dot line for dinaphthyl-methane, solid line for their sum.

$C_{18}H_{12}$ PAHs may be promising formation reactions of $C_{19}H_{14}$ PAHs. Methyl-triphenylene has a much higher mole fraction than methyl-chrysene and methyl-benzo[*c*]phenanthrene. Dash dot line in Fig. 4b presents the modeling results of methyl-triphenylene, which is quite a bit smaller than the experimental measurements. The reaction of naphth-1-yl-methyl and phenylacetylene and the combination of phenyl and phenalenyl may also contribute to $C_{19}H_{14}$, phenyl-phenalene, as shown by the dot line in Fig. 4b. However, the predicted mole fraction of phenyl-phenalene is even smaller than methyl-triphenylene, around one order of magnitude lower than the experimental measurements. H-elimination of phenyl-phenalene yields an RSR, phenyl-phenalenyl, which cyclodehydrogenizes to 4*H*-benzo[*cd*]fluoranthene. Fig. 4a presents good predictions of $C_{19}H_{12}$ PAH by this pathway.

Figs. 4c and 4d show the mole fractions of $C_{20}H_{14}$ and $C_{20}H_{12}$. 1,1-Binaphthalene and phenyl-phenanthrene are two molecular structures considered for $C_{20}H_{14}$, and the experimental mole fractions are calibrated as 1,1-binaphthalene. As discussed in Section 3.2.5, we neglect the chemistry of 1,2-binaphthalene in $\alpha$-methyl-naphthalene pyrolysis and merge its reaction flux into 1,1-binaphthalene. Binaphthalene may have a higher mole fraction than phenyl-phenanthrene. One might assume that it is because of the higher concentration of naphthalene than phenanthrene. However, kinetic analysis of the present model reveals dominant contributions from indenyl addition to naphth-1-yl-methyl and benzo-fulvenallene (~90% at both pressures). These two reactions are the additions of penta-ring RSRs with subsequent ring expansions, which have higher likelihood than aryl combination reactions. Cyclization and dehydrogenation of $C_{20}H_{14}$ PAHs fuse a new aromatic ring, yielding $C_{20}H_{12}$ PAHs, such as perylene and benzo[*j*]fluoranthene. The mole fraction of $C_{20}H_{12}$ is calibrated as perylene. Besides perylene, benzo[*j*]fluoranthene, benzo[*l*]acephenanthrylene, cyclopenta[*hi*]chrysene, benzo[*a*]pyrene, and benzo[*e*]pyrene are the $C_{20}H_{12}$ isomers included in the present model. The model predicted mole fractions of benzo[*a*]pyrene and benzo[*e*]pyrene are lower than other $C_{20}H_{12}$ isomers; therefore, their profiles are not presented in Fig. 4c. Benzo[*a*]pyrene and benzo[*e*]pyrene may be formed by the HAVA pathway of pyrene, but this pathway does not have a high efficiency in $\alpha$-methyl-naphthalene pyrolysis. Benzo[*j*]fluoranthene, benzo[*l*]acephenanthrylene, and cyclopenta[*hi*]chrysene have higher mole fractions than perylene. These three species are formed *via* the global reaction of benzo-fulvenallenyl and indenyl, as proposed in this study. It indicates that radical chain reactions of RSRs may have higher efficiencies for PAH formation than the dimerization and subsequent dehydrogenation of aryls. Although the reaction of indenyl and naphth-1-yl-methyl is also included for binaphthalene formation, further dehydrogenation process limits the conversion from binaphthalene to perylene and benzo[*j*]fluoranthene.

$C_{21}H_{16}$ PAHs may be methyl-binaphthalene and dinaphth-1-yl-methane. Modeling results of dinaphth-1-yl-methane show higher values than methyl-binaphthalene. Cyclization and dehydrogenation of $C_{21}H_{16}$ PAHs may fuse a new benzene ring, yielding $C_{21}H_{14}$ PAHs, such as 7*H*-benzo[*ij*]tetraphene and methyl-perylene. However, kinetic analysis reveals that the reaction of naphth-1-yl and benzo-fulvenallenyl provides a dominant contribution (>90% at both pressures). As shown in Fig. 4e, 7*H*-benzo[*ij*]tetraphene is the major $C_{21}H_{14}$ PAHs in $\alpha$-methyl-naphthalene pyrolysis.

Decomposition of $\alpha$-methyl-naphthalene mainly produces two $C_{11}H_9$ radicals, methyl-naphthyl and naphth-1-yl-methyl. The combination reactions of these two $C_{11}H_9$ radicals are critical in PAH growth. The reaction kinetics of $C_{22}H_{18}$ has been introduced in Part I of this study (SMM8). Fig. 5c presents the model predictions of $C_{22}H_{18}$. The sum of the predicted mole fractions of $C_{22}H_{18}$ isomers are at ~ $10^{-5}$ level, which is comparable to experimental measure-

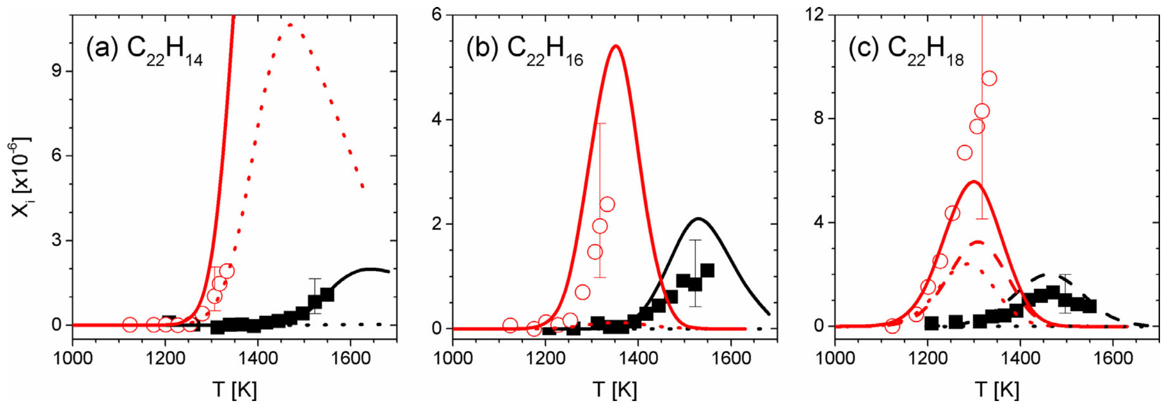



**Fig. 5.** Mole fraction profiles of $C_{22}$ PAHs measured in the experiment (scatter, EXP) and simulated by the present model (lines, MPR). (a) $C_{22}H_{14}$: solid line for picene, dot line for naphth-1-yl-acenaphthylene; (b) $C_{22}H_{16}$: solid line for dihydropicene, dot line for dinaphth-1-yl-ethylene; (c) $C_{22}H_{18}$: dash line for methyl-(naphth-1-yl-methyl)-naphthalene, dot line for dinaphth-1-yl-ethane, solid line for their sum.

ments. Dinaphth-1-yl-ethane ($A_2CH_2CH_2A_2$) is the combination product of two naphth-1-yl-methyl radicals, while methyl-(naphth-1-yl-methyl)-naphthalene ($A_2CH_2A_2CH_3$) represents the combination product of methyl-naphthyl and naphth-1-yl-methyl radicals. $A_2CH_2A_2CH_3$ is thermally more stable than $A_2CH_2CH_2A_2$ because benzyl C–C bond is stronger than the paraffinic C–C bond. $A_2CH_2A_2CH_3$ has a lower tendency of reverse bond fission reaction, compared to $A_2CH_2CH_2A_2$, and thus has a higher mole fraction.

Dehydrogenation of dinaphthyl-ethane yields dinaphthyl-ethylene; however, its mole fraction is much smaller than the experimental measurements. Dihydropicene, another $C_{22}H_{16}$ PAH considered in the present model, is formed by the reaction of naphth-1-yl-methyl and benzo-fulvenallenyl. Fig. 5b reveals the dominance of dihydropicene over dinaphthyl-ethylene. Subsequent dehydrogenation of dihydropicene yields 15% and 29% of picene at 30 and 760 Torr, respectively, as shown in Fig. 5a. Our model indicates that the recombination of two benzo-fulvenallenyl provides a comparable contribution to picene (>70%). Naphthyl-acenaphthylene is not important at all at 30 Torr, but its formation increases with the higher occurrence of H-abstraction reactions at 760 Torr.

### 4.2. Sensitivity analysis

Common simulation tools, like Chemkin-pro [70], usually support local sensitivity analysis at one specific temporal and spatial grid. We instead developed a Python code based on Cantera [53] to perform global sensitivity over the entire flow reactor (see details in SMM2). Sensitivity analysis of PAHs presented in this section are used to reveal critical reactions in the aromatic growth from $\alpha$-methyl-naphthalene pyrolysis. Figs. 6 and 7 present sensitivity analysis of several selected PAHs: acenaphthylene, phenanthrene, phenalenyl, pyrene, fluoranthene, benzo[*ghi*]fluoranthene, triphenylene, perylene, binaphthalene, 7*H*-benzo[*ij*]tetraphene, and picene. These selected species cover PAHs from bicyclic to pentacyclic aromatic structures.

Two reactions of the parent molecule, unimolecular C–H bond fission of $\alpha$-methyl-naphthalene and H-abstraction of $\alpha$-methyl-naphthalene by H-radical, are commonly sensitive to all species in Figs. 6 and 7 at 30 and 760 Torr, respectively. This finding suggests that the dominant fuel consumption pathway changes with the pressure, and naphth-1-yl-methyl radical provides the most significant contribution to PAH growth. The decomposition of benzo-fulvenallene to cyclopent-1-en-3–yne and benzyne is also sensitive to most of the selected PAHs in Figs. 6 and 7. The formations of acenaphthylene, phenanthrene, pyrene, fluoranthene, benzo[*ghi*]fluoranthene, triphenylene, and picene either show negative sensitivities to this reaction due to the competitiveness on benzo-fulvenallene or positive sensitivities to this reaction due to the formation pathways from cyclopent-1-en-3–yne or benzyne. The decomposition of naphth-1-yl-methyl to benzo-fulvenallene is another universally sensitive reaction, which impacts the carbon flux from $\alpha$-methyl-naphthalene to PAHs. Compared to the other $\alpha$-methyl-naphthalene derived radical, methyl-naphthyl, the sub-mechanism of naphth-1-yl-methyl plays a more significant role. H-abstraction reaction of $\alpha$-methyl-naphthalene by methyl radical has a smaller impact on naphth-1-yl-methyl formation than H-abstraction by H-atom. We observed the negative sensitivity of this methyl reaction to some PAHs, like acenaphthylene and pyrene, because the formation pathways of these species are highly dependent on methyl radicals; thus methyl radicals consumed by H-abstraction reactions have negative effects on them.

Previous studies [18,34,41,71] have recommended naphthyl to be a critical PAH precursor. Here, we have also found naphth-1-yl-methyl, benzo-fulvenallene, and benzo-fulvenallenyl to be key PAH precursors. We observed higher sensitivities of reactions involving RSRs, such as propargyl, fulvenallenyl, indenyl, and benzo-fulvenallenyl than traditional pathways of HACA, HAVA, and the dimerization of aryls.

In addition to those general sensitivities, each PAH has its own specific features. Acenaphthylene is very sensitive to the reaction of benzo-fulvenallenyl and methyl because it is one of its major contributors, particularly at low pressure. The reaction that converts acenaphthene to $A_2R_5H$ radical is also sensitive at 760 Torr, as shown in Fig. 6a.

Propargyl radical is a critical intermediate for phenanthrene. Recombination reaction between benzo-fulvenallenyl and propargyl forms phenanthrene, which certainly has a positive sensitivity. The decomposition of phenyl and the formation and subsequent decomposition of cyclopent-1-en-3–yne and benzyne are potential propargyl contributors, which consequently have positive sensitivities. The decomposition of propargyl to $C_3H_2$ diradical then shows a negative effect at 30 Torr.

Phenalenyl, the only PAH radical in sensitivity analysis, shows very different behavior at 30 and 760 Torr, as shown in Fig. 6c and d. At low pressure, phenalenyl formation relies on the combination of benzo-fulvenallene and other small intermediates. When pressure increases, the contribution from the decomposition of phenyl-phenalene becomes more and more important. Phenyl-phenalene works as an intermediate between naphth-1-yl, indenyl and phenalenyl radicals, which leads to a complicated carbon flux network. Therefore, the reactions which promote the formation of

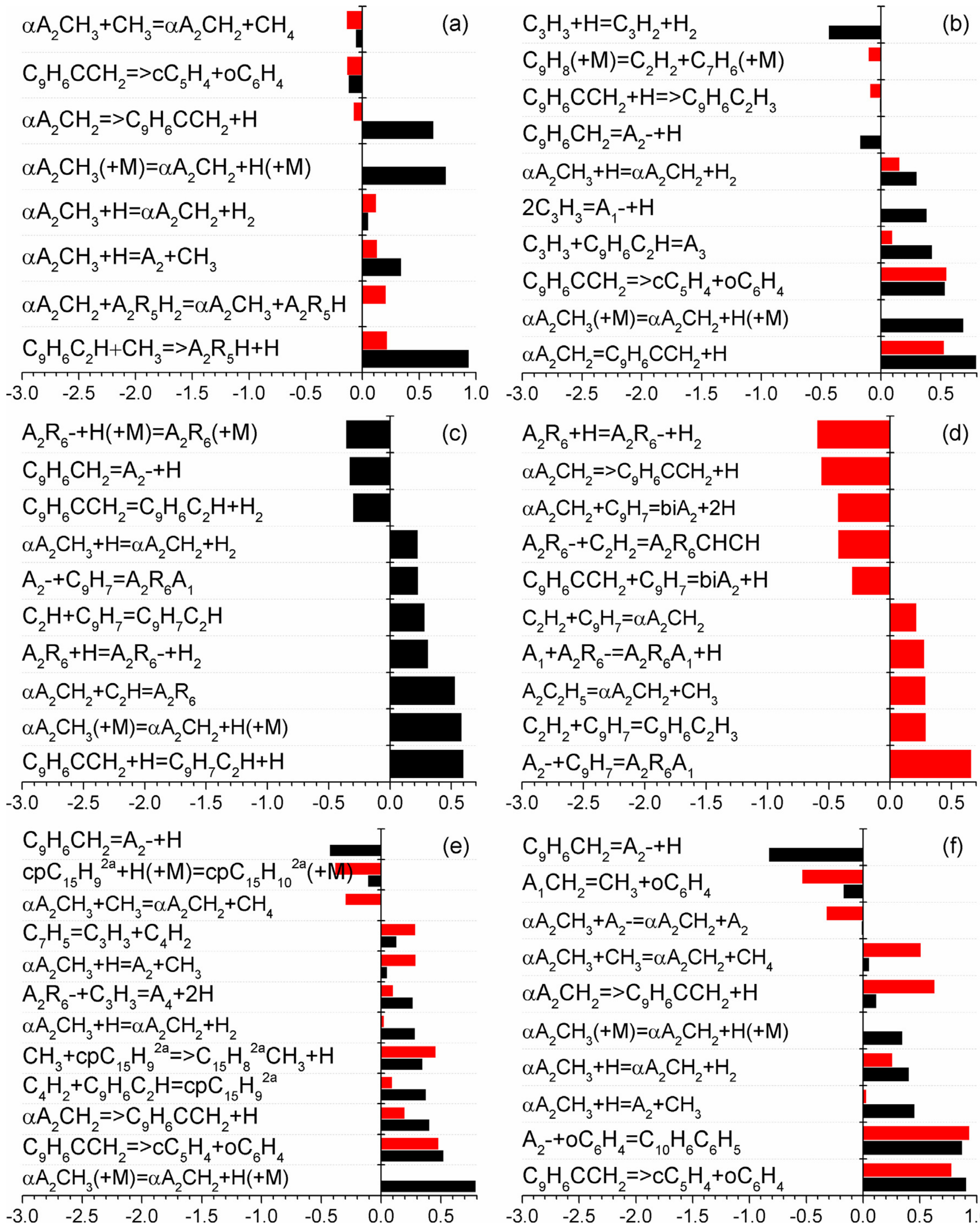



**Fig. 6.** Sensitivity analysis at 50% conversion of $\alpha$-methyl-naphthalene at 30 (black) and 760 Torr (red). (a) acenaphthylene, (b) phenanthrene, (c) and (d) phenalenyl at 30 and 760 Torr, (e) pyrene, (f) fluoranthene.

benzo-fulvenallene and indenyl have positive sensitivities and vice versa, e.g., the reaction of naphth-1-yl-methyl and indenyl for bi-naphthalene has negative sensitivity to phenalenyl.

The sensitivity analysis of pyrene shows that methyl addition to $cpC_{15}H_9$–2a radical and subsequent ring expansion may have a considerable contribution to pyrene formation, especially considering that the HACA pathway for pyrene from phenanthrene may be overestimated [58,72,73]. The reaction between phenalenyl and propargyl is also significant for pyrene [74], as shown in Fig. 6e. The other $C_{16}H_{10}$ PAH, fluoranthene, has a completely different chemistry from pyrene [19,42]. Sinha et al. [42] and Zhao et al. [19] recommended the critical role of phenyl-naphthyl radical in the formation of fluoranthene, derived from benzyl and phenyl reactions, respectively. We instead propose a reaction route for phenyl-naphthyl *via* the reaction of benzyne and naphthyl. Fig. 6f shows its sensitivity at both pressures, as well as benzyne promoting and consuming reactions.

For benzo[*ghi*]fluoranthene, we propose a formation route by the combination of two RSRs, fulvenallenyl and benzo-fulvenallenyl, besides the traditional pathway of the dehydrogena-

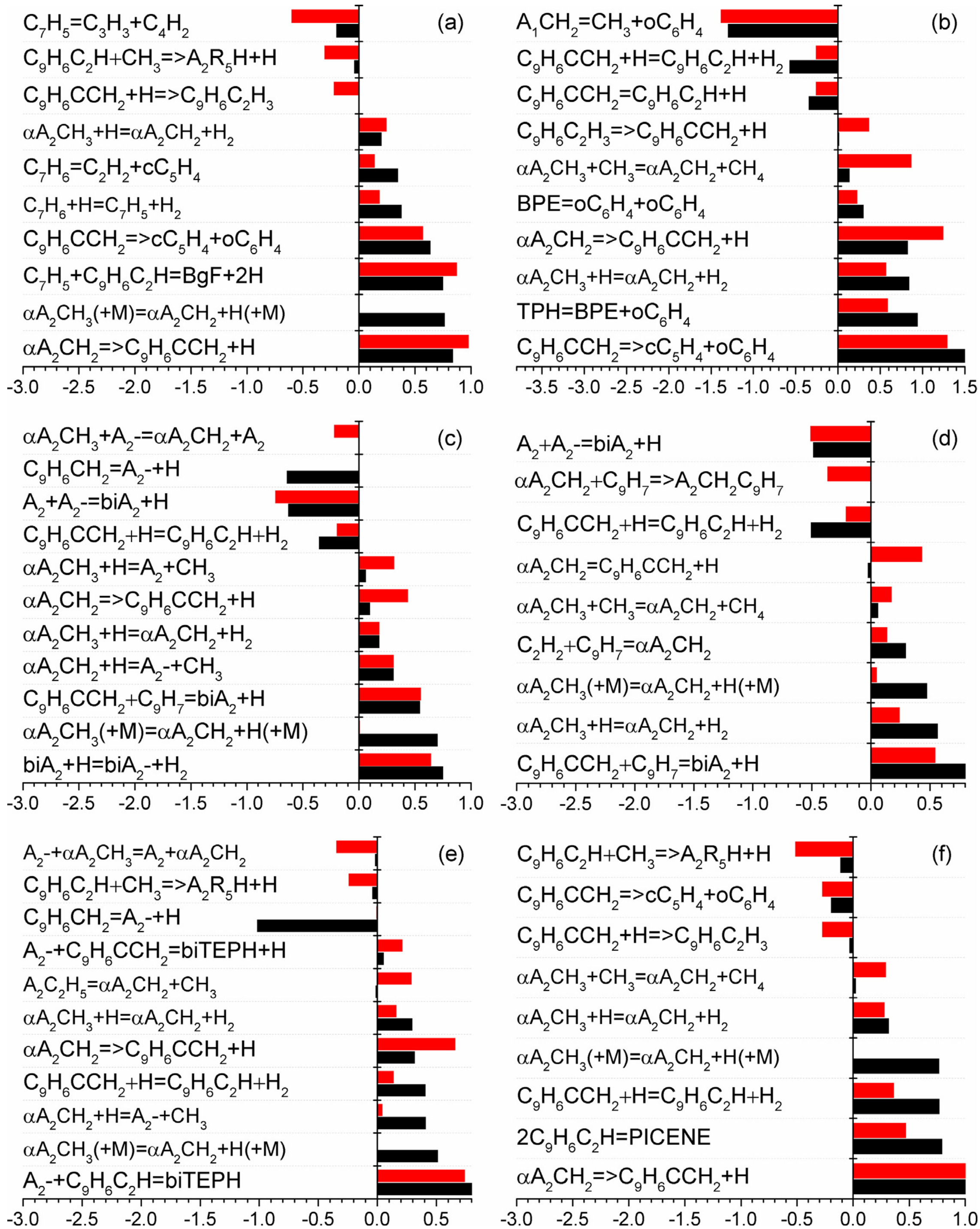



**Fig. 7.** Sensitivity analysis at 50% conversion of $\alpha$-methyl-naphthalene at 30 (black) and 760 Torr (red). (a) benzo[*ghi*]fluoranthene, (b) triphenylene, (c) perylene, (d) binaphthalene, (e) 7*H*-benzo[*ij*]tetraphene, (f) picene.

tion of benzo[*c*]phenanthrene. This reaction not only improves the model prediction of benzo[*ghi*]fluoranthene (Fig. 3b) but also is very sensitive at both pressures (Fig. 7a). The reactions related to the formation and consumption of these two RSRs reveal positive and negative sensitivities, respectively. The decomposition of benzo-fulvenallenyl to benzyne and cyclopent-1-en-3–yne also shows strong positive sensitivity because benzyne is the main precursor for benzyl radical which in turn promotes fulvenallenyl formation.

Triphenylene (Fig. 7b) is closely related to benzyne, while its $C_{18}H_{12}$ isomers, benzo[*c*]phenanthrene and chrysene, are closely related to indenyl [17]. The reaction of biphenylene and benzyne, which is proposed as the formation pathway of triphenylene in the present model, shows positive sensitivity, as well as the combination reaction of two benzynes forming biphenylene.

Besides the traditional aryl dimerization reactions, we propose radical chain reactions of RSRs for binaphthalene in the present model. Further dehydrogenation of binaphthalene makes perylene.

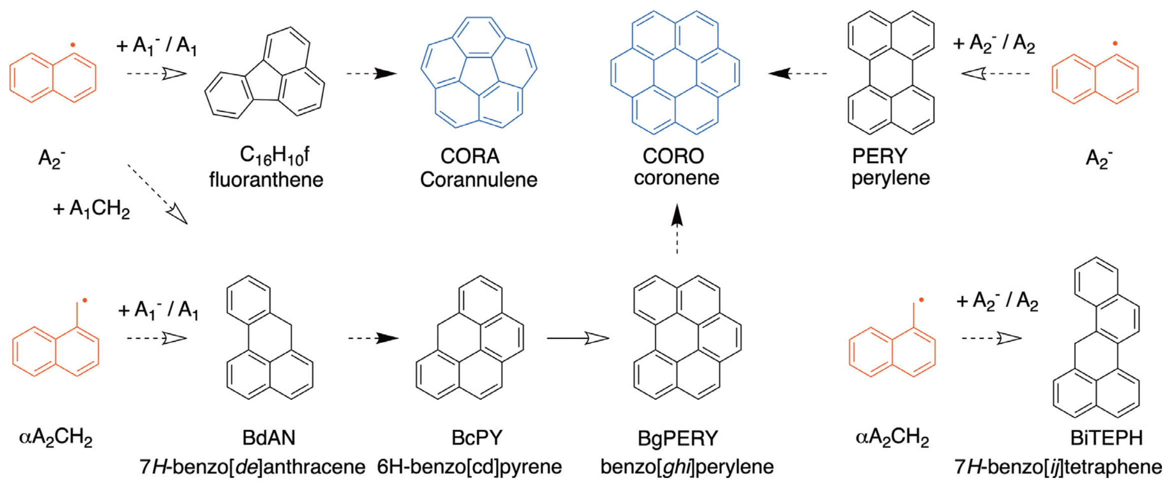



**Fig. 8.** Route 1: addition and cyclization reaction sequences of naphth-1-yl-methyl and naphth-1-yl radicals. Dash hollow arrows indicate Route 1, dash solid arrows indicate acetylene addition reaction, solid hollow arrows indicate Route 3.

H-abstraction of binaphthalene is the most sensitive reaction for perylene. As shown in Figs. 7c and d, the reaction of indenyl addition to benzo-fulvenallene is very sensitive to the formation of perylene and binaphthalene. Decomposition reaction of indenyl-methyl forming naphth-1-yl and H-abstraction reaction of benzo-fulvenallene have negative sensitivities, while the decomposition of naphth-1-yl-methyl forming benzo-fulvenallene has positive sensitivity. It is also interesting to see that naphth-1-yl addition to naphthalene has a negative sensitivity. This may be because the radical chain reaction of indenyl is very efficient to form binaphthalene, so the reaction of naphth-1-yl and naphthalene acts in a reverse way consuming binaphthalene.

Figs. 7e and f indicate that benzo-fulvenallenyl is critical for benzo[*ij*]tetraphene and picene because this radical is involved in the main formation pathways of these two PAHs. Reactions that contribute to the sequences of naphth-1-yl-methyl → benzo-fulvenallene → benzo-fulvenallenyl have positive sensitivities. Naphth-1-yl radical is another important precursor of benzo[*ij*]tetraphene. Ipso-reaction of naphth-1-yl-methyl leading to naphth-1-yl has positive sensitivity, while H-abstraction by naphth-1-yl has negative sensitivity. Our analyses on benzo[*ij*]tetraphene and picene in this study are limited by our current understandings of the chemistry of large PAHs. There may be other efficient formation pathways for these two PAHs that should be explored in future works.

### 4.3. General picture of aromatic growth

Based on the kinetic analysis above, we found several prototypical PAH formation pathways, besides traditional HACA and HAVA mechanisms, in the pyrolysis of $\alpha$-methyl-naphthalene. All these pathways are intertwined and overlap in the PAH growth process. Addition and cyclization reactions of naphth-1-yl-methyl and naphth-1-yl (Route 1 shown in Fig. 8) provide considerable contributions to PAH formation from $\alpha$-methyl-naphthalene. Compared to HACA and HAVA mechanisms, Route 1 shows an obvious advantage in the formation of tetracyclic and pentacyclic PAHs. Zhao et al. [19] proposed the mechanism of phenyl addition and dehydrocyclization for PAH growth. Based on our kinetic study on $\alpha$-methyl-naphthalene pyrolysis, this mechanism may be extended to naphth-1-yl, naphth-1-yl-methyl and other radicals of peri–condensed aromatic hydrocarbons. As shown in Fig. 8, such combination and cyclization reaction sequences are also efficient between naphth-1-yl-methyl and naphthalene/naphthyl. The addition of benzyl-like radicals to peri–condensed aromatic hydrocarbons and their radicals yields PAHs with phenalene structure in their molecules. Benzyne may also undergo such reaction sequences which show higher efficiencies for PAH growth in the present model.

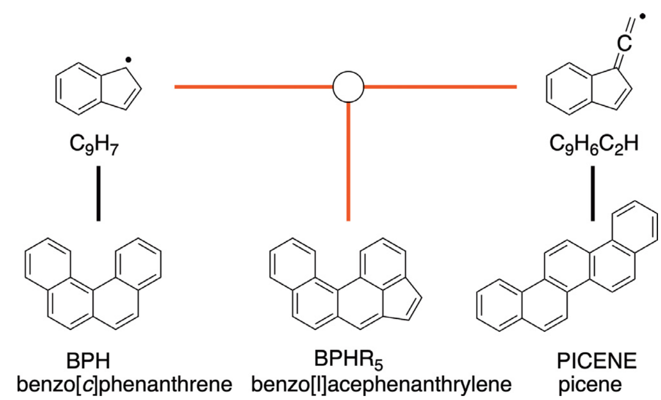



**Fig. 9.** Route 2: recombination of RSRs and the subsequent ring expansion. Black lines indicate self-combination reactions, red lines indicate cross-combination reactions of two RSRs.

Recombination of RSRs and the subsequent ring expansion is the 2nd prototypical pathway (Route 2 shown in Fig. 9). RSRs with cyclopentadienyl in the molecule structures may undergo ring expansion after combining with other intermediates, proceeding to benzene rings [60]. This reaction sequence may also be applied to RSRs with fulvenallenyl structure, as shown in Fig. 9. The self-combination reactions of indenyl and benzo-fulvenallenyl lead to benzo[*c*]phenanthrene and picene, respectively. The combination reaction of indenyl and benzo-fulvenallenyl forms benzo[*l*]acephenanthrylene. Indeed, each pair of RSRs has the chance to undergo Route 2, similar to those shown in Fig. 9, unless these two radicals have very different residence times in a flame. Large ROP contributions and sensitivities to the formation of perylene, binaphthalene and picene confirm the efficiencies of the Route 2 pathways proposed in the present model.

Sequential propargyl addition is the proposed Route 3 for PAH formation from naphth-1-yl radical. As shown in Fig. 10, PAHs with peri–condensed and phenalene-like structures are alternately formed in this reaction sequence. We observed and quantified phenalene and pyrene in this study, and confirmed the contribution of Route 3 in pyrene formation. Phenalenyl, which is another kind of RSR with a six-member ring, is the critical intermediate in

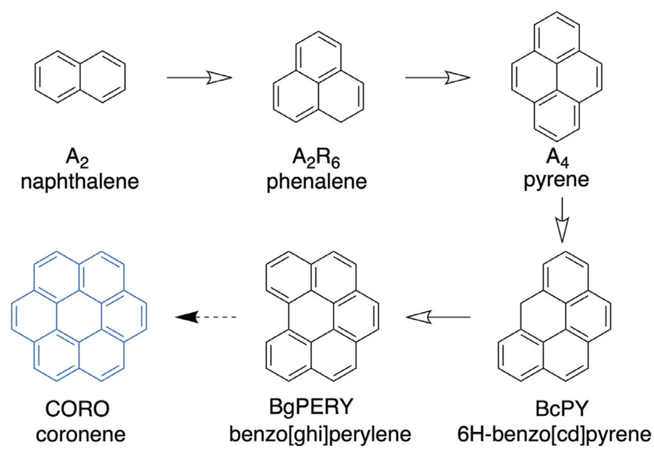



**Fig. 10.** Route 3: sequential propargyl addition pathway starting from naphthalene. Hollow arrows indicate propargyl addition reaction, solid arrow indicates acetylene addition reaction.

this process. Phenalene is formed by propargyl addition over the *zigzag* periphery of naphthalene; further C–H bond fission or H-abstraction convert phenalene to a fully resonance radical. Another propargyl addition then turns phenalenyl back to a peri–condensed aromatic structure. Phenalenyl radical represents a class of radicals; for example, H-abstraction reaction of 6*H*-benzo[*cd*]pyrene may form another phenalenyl-like RSR. Further propargyl addition to 6*H*-benzo[*cd*]pyrenyl produces benzo[*ghi*]perylene. Phenalenyl-like RSR can also be formed *via* acetylene addition to benzyl-like RSR, and, therefore, sequential additions of methyl and acetylene to aryl radicals has the identical result as propargyl addition reaction.

## 5. Conclusions and perspectives

PAH formation chemistry was investigated during $\alpha$-methyl-naphthalene pyrolysis in a flow reactor at 30 and 760 Torr over the temperature range of 1000 - 1550 K using SVUV-PIMS. A series of PAHs, from bicyclic to pentacyclic aromatic hydrocarbons, were quantified as well as some important free radicals. Mole fraction profiles were measured as a function of reactor temperature. A PAH kinetic model was developed based on the previous indene model [17]. We analogized the chemistry of naphthyl, benzo-fulvenallene, and benzo-fulvenallenyl, derived from the decomposition of $\alpha$-methyl-naphthalene, to the chemistry of benzyl, fulvenallene, and fulvenallenyl. We also postulated new PAH formation pathways, particularly radical chain reaction between RSRs, such as indenyl, benzo-fulvenallenyl, naphth-1-yl-methyl, and phenalenyl. The proposed model exhibits good reproduction of the current flow reactor experimental data.

By performing ROP and sensitivity analysis, critical PAH formation pathways were identified for $\alpha$-methyl-naphthalene pyrolysis, such as: 1) addition and cyclization reactions of naphth-1-yl-methyl and naphthyl, 2) radical chain reactions of RSRs and subsequent ring expansion reactions, 3) sequential propargyl addition reactions. In general, two $\alpha$-methyl-naphthalene related reactions, unimolecular C–H bond fission and H-abstraction by H-radical, are sensitive to the PAH formation processes. The sequential decomposition of naphth-1-yl-methyl to benzo-fulvenallene and then to benzo-fulvenallenyl controls the carbon flux from $\alpha$-methyl-naphthalene to PAHs. The decomposition of benzo-fulvenallene to cyclopent-1-en-3–yne and benzyne, as well as benzyne addition and cyclization, are quite important in this reaction system, particularly in the formation of fluoranthene and triphenylene. We also found higher efficiencies of RSR reactions, such as those of propargyl, fulvenallenyl, indenyl, and benzo-fulvenallenyl, than traditional pathways of HACA, HAVA, and the dimerization of aryls.

This study might challenge our conventional understandings of PAH and soot formation, which for quite a long time has been limited to the addition reactions of small intermediates. The combination and aromatization reactions of aromatics with moderate molecular sizes, not only aryl radicals but also aromatic RSRs, reveal considerable contributions to PAH growth. This finding infers a coalescence of soot by the reactions of very large PAHs which may benefit the future development of soot models not only from peri–condensed aromatics but also from RSRs and side-chain substituted PAHs. Further quantum chemical and experimental studies are required to validate these prototypical elementary reactions.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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## Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2021.111530.

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